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Political Discourse on and Economic Impact of Migration: An Empirical Analysis

Masterthesis

presented by

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List of Abbreviations

NLP	natural language processing
PCA	Principal Component Analysis
XML	Extensible Markup Language
TF-IDF	Term Frequency-Inverse Document Frequency
PV-DBOW	Paragraph Vector - Distributed Bag of Words
LLM	large language models
OECD	Organisation for Economic Co-operation and Development

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1 Introduction

"We have to stop the immigrant invasion!" Such rhetoric, deliberately provocative as it may be, has become increasingly familiar in contemporary political debate. Over the past two decades, immigration has gained growing prominence in political and policy debates across several advanced democracies (Ivarsflaten 2008). Its salience stems not only from the direct economic and social consequences of migration but also from its capacity to serve as a focal point through which broader concerns about employment, demographic change, public services, and social cohesion are articulated. In particular, immigration has become an important issue through which right-wing populist parties mobilize broader economic and social grievances (Ivarsflaten 2008; Fang and Speer n.d.).

Political actors, however, frequently invoke similar underlying economic conditions to justify fundamentally different positions on immigration. In South Africa, for example, the anti-immigrant movement Operation Dudula has accused immigrants of taking jobs from citizens against the backdrop of persistently high unemployment (Yu 2025; Hlatshwayo 2023). In France, the Reconquête party has employed the "great replacement" narrative, linking immigration to declining birth rates and framing demographic change as a threat (Ekman 2022). In Germany, the AfD has adopted predominantly restrictive positions on immigration, pointing to the neglect of domestic education, vocational training, and technological automation in favor of relying on a lower-cost foreign workforce (Deutscher Bundestag 2023b). In the United Kingdom, Reform UK has linked immigration to pressures in the housing market and advocated restrictions on access to social housing for foreign nationals (Reform UK 2025). Conversely, political actors supporting immigration frequently emphasize its potential economic benefits, including alleviating labour shortages, contributing to economic output, and mitigating the consequences of population ageing (Ekman 2022). Thus, the same economic conditions—including unemployment, demographic ageing, labour shortages, and housing pressures—can be interpreted and mobilized to support opposing political narratives.

This tension raises a question that extends beyond the established relationship between economic conditions and political discourse: whether the relationship may also operate in the opposite direction. Existing research documents substantial macroe-

conomic effects of migration, including effects on employment, wages, consumption, investment, and output (Maffei-Faccioli and Vella 2021; Ortega and Peri 2009; Abramitzky and Boustan 2017). A large literature likewise examines how economic conditions shape public and political attitudes toward immigration (Abdelatif and Andriantomanga 2026; Scheve and Slaughter 2001; Hainmueller and Hopkins 2014; Dustmann and Preston 2007). Comparatively less attention, however, has been devoted to whether immigration-related political discourse itself contains information about subsequent migration and macroeconomic dynamics. Such a relationship is plausible because political discourse may contribute to shaping expectations about future policy, economic conditions, and migration before these expectations are reflected in realized outcomes. At the same time, economic and migration developments may feed back into political discourse, creating a dynamic relationship in which neither direction can be considered in isolation.

This thesis investigates whether immigration-related political sentiment and macroeconomic conditions are dynamically related, allowing for feedback in both directions rather than imposing a one-directional relationship a priori. To this end, we extend the Bayesian Vector Autoregression (BVAR) framework of (Maffei-Faccioli and Vella 2021), which examines the macroeconomic effects of migration shocks in Germany, by incorporating a novel text-based measure of immigration sentiment derived from speeches in the German Bundestag. The sentiment measure is constructed using a purpose-built dictionary designed to distinguish between restrictive and welcoming positions toward immigration and is subsequently incorporated into the VAR system as an endogenous variable alongside migration and macroeconomic indicators. This framework makes it possible to examine whether changes in immigration-related political sentiment contain predictive information about subsequent developments in migration and macroeconomic conditions, while simultaneously accounting for the possibility that economic and migration developments influence subsequent political sentiment.

The main methodological contribution of this thesis is therefore the integration of parliamentary text data into a multivariate macroeconomic time-series framework. Rather than treating political discourse solely as an outcome of economic and migration conditions, the analysis treats sentiment expressed in parliamentary debates as a measurable time-varying variable that may contain information about future developments. This approach combines quantitative text analysis with Bayesian time-series

modelling and provides a framework for studying the temporal interaction between political discourse, migration, and macroeconomic conditions. Importantly, the empirical framework does not presume that political sentiment causes subsequent economic or migration outcomes. Instead, it allows the temporal structure of the data to reveal whether systematic dynamic relationships and predictive interactions exist between these variables.

Germany provides a particularly suitable setting for this analysis. First, the study can build directly on the empirical framework developed by (Maffei-Faccioli and Vella 2021), allowing the analysis to extend an established examination of migration and macroeconomic dynamics. Second, Germany has experienced substantial variation in both migration flows and the political salience of immigration. The 2015–2016 migration episode represented a major turning point in German immigration politics and generated extensive parliamentary debate. More recently, the large-scale displacement from Ukraine in 2022 produced another substantial increase in migration and renewed political attention to immigration. These episodes, together with longer-term changes in migration and economic conditions, provide substantial temporal variation for examining whether shifts in immigration-related political sentiment coincide with, precede, or follow changes in migration and macroeconomic indicators.

The remainder of this thesis is organized as follows. Section 2 reviews the relevant literature on migration, macroeconomic dynamics, political attitudes, and political discourse. Section 3 presents the parliamentary text data and describes the data preparation procedure, the construction and refinement of the immigration sentiment dictionary, the construction of the sentiment measure from Bundestag speeches, and its validation. Section 4 introduces the Bayesian VAR framework, presents the macroeconomic and migration variables used in the analysis, discusses the identification strategy, presents and discusses the empirical results together with a series of robustness checks. Section 5 concludes.

2 Literature Review

This thesis builds on three strands of literature: research on the macroeconomic effects of migration, research on how economic conditions shape attitudes and political rhetoric toward immigration, and research that turns political text into quantitative in-

dicators. Each strand covers one link in the relationship studied here. None of them, however, models these links jointly.

Migration has long been closely tied to economic conditions. In the push–pull framework, unemployment, poverty, conflict, and political instability push people to leave their home countries, while better economic and social conditions attract them to destination countries (Lee 1966; Hatton and Williamson 2005). Destination states have, in turn, used migration policy to address labour shortages and demographic change, and today such policy is closely linked to labour demand, security, integration, and migrants' rights (Hollifield 2004). Ageing populations and persistent labour shortages continue to raise the demand for foreign workers in advanced economies (OECD 2024b; OECD 2024a). Regarding the consequences of migration, Franz, Oser, and Winker (1994) show in a macroeconometric disequilibrium model that the effects of migration into West Germany depend on the prevailing economic conditions and market regimes. More recently, Maffei-Faccioli and Vella (2021) use a structural VAR for Germany and find that net migration shocks affect vacancies, wages, house prices, consumption, investment, net exports, output, and unemployment. Their framework is the starting point of the empirical analysis in this thesis. Like most of this literature, however, it treats realised migration flows as the driving force and does not ask whether the political debate surrounding migration carries information of its own.

A second strand examines the reverse direction: how economic and social conditions shape attitudes toward immigration. Survey-based studies relate opposition to immigration to labour-market competition, fiscal concerns, and cultural considerations (Scheve and Slaughter 2001; Dustmann and Preston 2007; Hainmueller and Hopkins 2014). Concerns about public services, housing, security, and cultural identity add to these economic motives (Huysmans 2000; Young, Loebach, and Korinek 2018; OECD 2024a), and threat-based messages can amplify them (Brader, Valentino, and Suhay 2008; Esses, Jackson, and Armstrong 1998). Political actors respond to and shape these concerns. Right-wing populist parties in particular mobilise broader economic and social grievances through the immigration issue (Ivarsflaten 2008). Analysing 140 years of US political speeches, Card et al. (2022) show that the framing of immigration has become more positive on average but increasingly polarised between parties.

Germany illustrates how strongly immigration discourse can shift with political and economic conditions. The Gastarbeiter agreements of the 1950s and 1960s framed im-

migration mainly as a response to labour shortages. In the early 1990s, rising asylum applications coincided with more restrictive rhetoric and the 1993 asylum compromise, while reforms around 2000, including birthright citizenship and the work of the Süssmuth Commission, marked a more open phase (Borkert and Bosswick 2007). The 2015–2016 refugee inflow transformed the political debate once again, whereas the arrival of refugees from Ukraine in 2022 met with broad institutional and public support (Trauner and Turton 2017; Brücker 2022).

The study closest to this thesis is Abdelatif and Andriantomanga (2026). They construct a migration sentiment index from Bundestag speeches for 1991–2018, using openness and restriction terms around migration anchors with a correction for negation. They find that higher economic and migration-related uncertainty is followed by more restrictive parliamentary rhetoric. Their design treats rhetoric as the outcome variable, so it captures only one direction of the relationship. In addition, their sample ends in 2018. It therefore excludes the Ukrainian displacement of 2022 and most of the period in which the AfD has been represented in the Bundestag.

The digitisation of parliamentary records has made political discourse measurable at scale, but automated measures are only as informative as their validation (Grimmer and Stewart 2013). Dictionary methods are transparent and efficient, yet they can miss context, negation, and reported speech (Kostikova et al. 2024). Supervised models and large language models capture more context, but they require more computation, often rely on manual labelling, and are harder to reproduce. Kostikova et al. (2024), for example, use large language models to measure solidarity and anti-solidarity toward migrants in Reichstag and Bundestag speeches over 155 years. Applying computational methods to UK parliamentary debates, Ghafouri et al. (2025) document a shift toward securitised and border-control framing and a declining emphasis on integration. This trade-off between transparency and contextual accuracy motivates the measurement strategy of this thesis. The main measure is built from a manually seeded dictionary that is expanded with word embeddings trained on the corpus itself, so that it adapts to parliamentary language while remaining interpretable. As a robustness check, immigration sentiment is additionally estimated with GPT-4.1-nano, following Kostikova et al. (2024).

A growing literature shows that sentiment extracted from parliamentary text contains economically relevant information. Latifi et al. (2024) construct a fiscal-policy senti-

ment measure from Bundestag speeches, include it in a recursively identified VAR, and find that changes in fiscal sentiment are followed by changes in government spending and other macroeconomic variables. Tillmann (2025) shows that monetary tightening affects fiscal sentiment differently for government and opposition speakers. Together, these studies suggest that parliamentary text can enter macroeconomic models as an endogenous variable, and that the speaker's political position matters for interpreting it. To my knowledge, this approach has not yet been applied to immigration.

Taken together, the literature establishes three relationships. Migration affects economic outcomes, economic and migration conditions shape immigration attitudes and parliamentary rhetoric, and parliamentary text can contain measurable economic information. What remains open is whether immigration sentiment expressed in parliament contains information about subsequent migration and macroeconomic dynamics, or whether it primarily reflects conditions that are already underway. This thesis addresses that question in three ways. First, it extends the BVAR framework of Maffei-Faccioli and Vella (2021) by an immigration sentiment measure treated as an endogenous variable, which allows feedback in both directions instead of imposing one a priori. Second, following Tillmann (2025), it distinguishes between government and opposition speakers. Third, it extends the sample to 2025 and tests robustness across sample periods, recursive orderings, and an independent LLM-based sentiment measure. The thesis thereby connects migration economics with computational political text analysis.

3 Text data

3.1 Data Sources and Corpus Construction

We use a compiled dataset of speeches from the German Bundestag, provided by Department 02: Business Studies and Economics at Justus-Liebig-Universität Gießen. The dataset comes from parliamentary speeches available as digital Extensible Markup Language (XML) files. These raw files are not ready for natural language processing (NLP) analysis and need extraction and preprocessing first.

Our corpus combines two sub-datasets. Both end up with the same structure, but they were extracted differently, since the Bundestag changed its XML format over

time. Table 1 shows both sub-datasets.

Table 1: Overview of the two sub-datasets underlying the combined corpus.

	Sub-dataset 1	**Sub-dataset 2**
Coverage	Q1 1960 – Oct. 2017	Oct. 2017 – Q2 2025
XML structure	Unstructured (continuous debate text)	Structured (individual speeches)
Extraction tool	Custom workflow, Latifi et al. (2024)	`pybundestag` package

The first sub-dataset covers the first quarter of 1960 to October 2017. In this period, the XML files store whole debates as continuous text, with only limited metadata. To turn these into speech-level data, Latifi et al. (2024) built a custom extraction workflow, first designed for proceedings from 7 September 1949 to 7 September 2021. Each extracted speech is linked to its session and date, and includes the speaker's role and party.

The second sub-dataset covers October 2017 to the second quarter of 2025. After a format change in the 19th legislative period, records became structured, so speeches and their metadata could be extracted directly. For this period, Latifi et al. (2024) used the `pybundestag` package. The result matches the structure of the first sub-dataset. Combining both gives one consistent speech-level corpus from Q1 1960 to Q2 2025.

Preprocessing was done by the data provider, following Latifi et al. (2024); we received the corpus already cleaned. The steps were: (i) removing irrelevant speeches; (ii) lemmatization; (iii) normalizing German umlauts and the Eszett (ß); (iv) removing line breaks, digits, and special characters; (v) lowercasing all text; (vi) removing single-character tokens and tokens longer than 30 characters; and (vii) removing stop words.[1]

This pipeline produces a final dataset of exactly 260,660 parliamentary speeches, covering Q1 1960 to Q2 2025.

3.2 Migration Speech Identification and Dictionary Construction

To make sentiment scores easier to interpret, and to reduce computing time, we keep only speeches that clearly discuss immigration. We identify these using keyword filtering, based on the keyword list from Kostikova et al. (2024). Since our filter uses

[1] Stop words are very common words that carry little meaning and add little to text analysis.

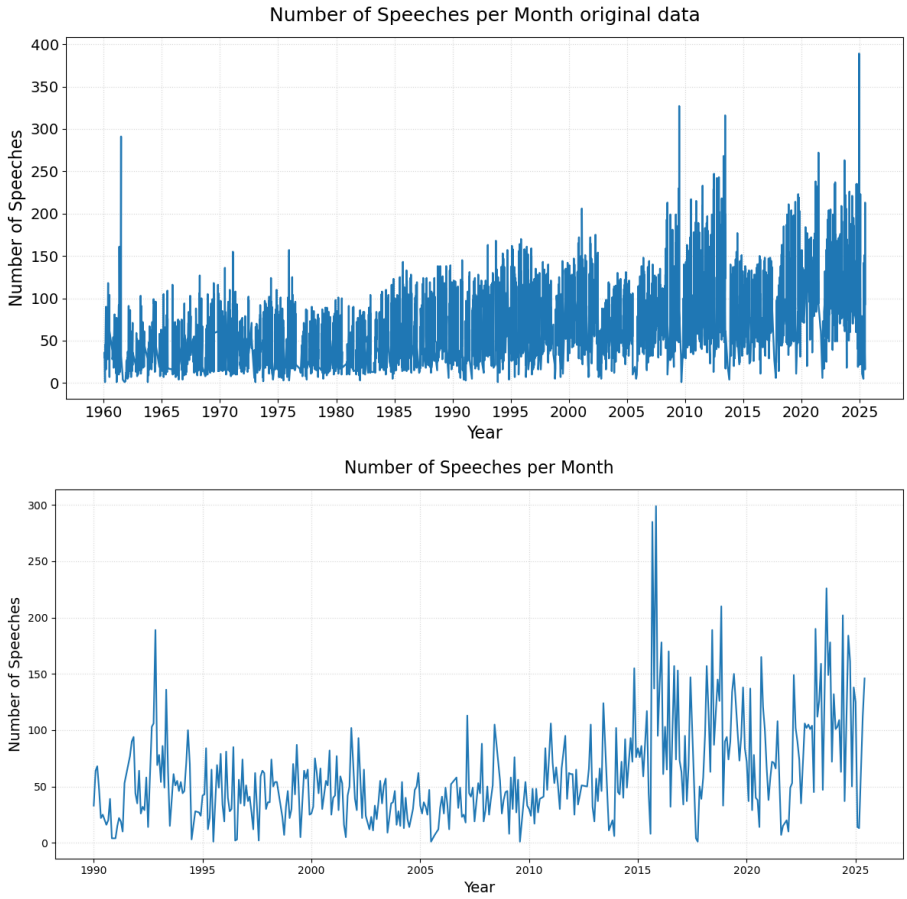


Figure 1: Monthly number of parliamentary speeches (top: original data; bottom: immigration-related speeches after keyword filtering).

Python's `re` module for pattern matching, we merge word forms that share the same root (e.g., *Flüchtling*, *Flüchtlinge*, *Flüchtlingen*) into a single stem, so no two entries overlap. This reduces the original list of 32 word forms (Table 2) to 13 stems covering the same meaning without duplication (Table 3).

This filtering reduces the corpus from 260,660 to 28,662 speeches, about 11% of all speeches. Figure 1 shows the counts before and after filtering. While total speeches have grown steadily over time, immigration-related speeches have grown even more, especially during periods of high migration.

Notable peaks are in the years 1993 and 2016, corresponding to serious tension around migration politics. After 2016 and an additional event (immigration from Ukraine), debates around that topic are more present, fluctuating around an average of 100 speeches per month.

From the reduced corpus, we build two sentiment dictionaries: a welcoming (open) dictionary and a restrictive dictionary. We build them in three steps: manual seed

Table 2: Original keyword list (32 inflected word forms), following Kostikova et al. (2024).

Keywords			
Flüchtlinge	Ausländer	Flüchtlingen	Zuwanderung
Vertriebenen	Ausländern	Asylbewerber	Migranten
Migration	Heimatvertriebenen	Aussiedler	Einwanderung
Ansiedler	Vertriebene	Zuwanderer	Asylbewerbern
Flüchtling	Heimatvertriebene	Sowjetzonenflüchtlinge	Aussiedlern
Einwanderer	Asylsuchenden	Asylsuchende	Bürgerkriegsflüchtlinge
Zuwanderern	Ansiedlern	Migrantinnen	Vertriebener
Emigranten	Kriegsflüchtlinge	Ausländerinnen	Immigranten

Table 3: Reduced list of 13 word stems used for regular-expression matching. Word forms sharing a root were merged into one stem, so no entry overlaps with another.

Word stems			
`migration`	`zuwander`	`einwander`	`asyl`
`fluechtling`	`gefluechtet`	`schutzsuchend`	`auslaend`
`vertrieben`	`migrant`	`aussiedler`	`ansiedler`
`buergerkriegs`			

selection, Word2Vec-based expansion, and manual cleaning.

We manually pick speeches that reflect different immigration positions, based on the documented history of German parties on this issue: 200 AfD speeches from 2017–2025, 94 CDU/CSU speeches from 1990–1994, and 500 CDU/CSU, SPD, and FDP speeches from the 2015–2018 refugee period. Close reading of these speeches gives us 655 seed words for the welcoming dictionary and 872 for the restrictive dictionary.

We represent our speeches as a unique text corpus and use the word2vec method developed in Mikolov et al. (2013) to represent words as vectors, so that we can capture relationships between them based on their numerical representation. We train a skip-gram Word2Vec model on the immigration-related corpus (vector size 100, context window 7, minimum word count 10, 100 training epochs, fixed random seed). We expand the dictionaries iteratively: in each round, we use a small, topic-focused seed list (for example, terms about the economic side of migration policy). For each seed word in the model's vocabulary, we find its 20 nearest neighbors by cosine similarity. We keep a candidate word only if (i) its similarity to the seed is at least 0.72, (ii) it appears in a Term Frequency-Inverse Document Frequency (TF-IDF) vocabulary built on the immigration corpus ($max_features$ = 80,000, min_df = 2), and (iii) it is a purely alphabetic token longer than three characters. Words that appear as neighbors of seeds

from both dictionaries are treated as ambiguous and dropped from both. We merge the kept words from each round into the growing dictionaries, ending with 755 welcoming words and 968 restrictive words.

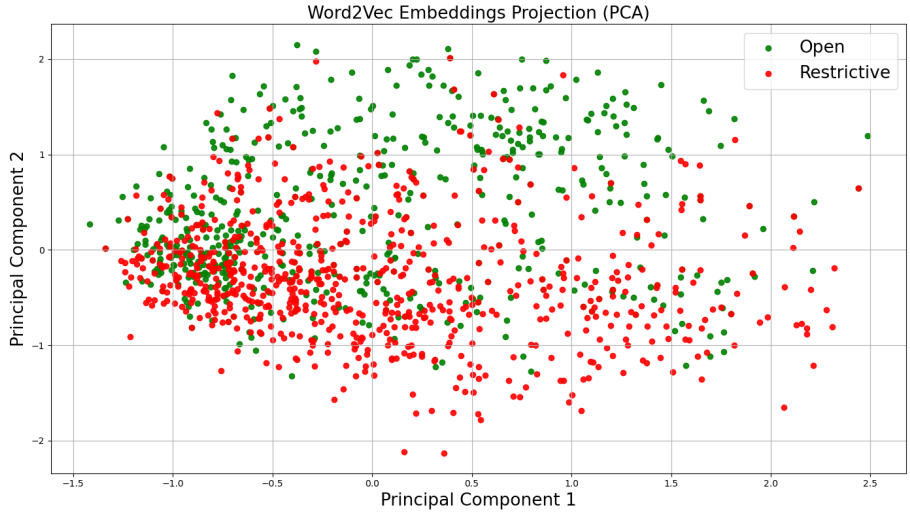


Figure 2: PCA projection of the expanded immigration dictionaries before cleaning (restrictive = red, welcoming = green).

To have a visual representation of our dictionaries, we project the word embeddings onto two dimensions using Principal Component Analysis (PCA), which keeps most of the variance while giving an interpretable view of the embedding space. As Figure 2 shows, restrictive terms (red) mostly sit lower in the projection, while welcoming terms (green) sit higher. By visual analysis of the graph, the interval $[-0.48, 0.5]$ on the y-axis holds most ambiguous words. To sharpen the split between the two dictionaries, we keep in our respective dictionaries only the restrictive words with a y-coordinate below -0.48 and the welcoming words with a y-coordinate above 0.5, dropping everything in between, which increases the semantic difference between the dictionaries. This gives cleaner dictionaries (Figure 3) of 288 restrictive and 241 welcoming terms.

We then manually clean the dictionaries, prompted by an early check of average sentiment scores by party (Section 3.3). This check found unusual negative scores for parties known to hold welcoming positions, especially Bündnis 90/Die Grünen and Die Linke. For each of these parties, we randomly sampled speeches with a sentiment score below zero and read them individually.

This revealed two problems. First, restrictive words were sometimes used critically, for example when a speaker quoted or condemned anti-immigration rhetoric rather than expressing it themselves — a nuance our embedding-based measure can-

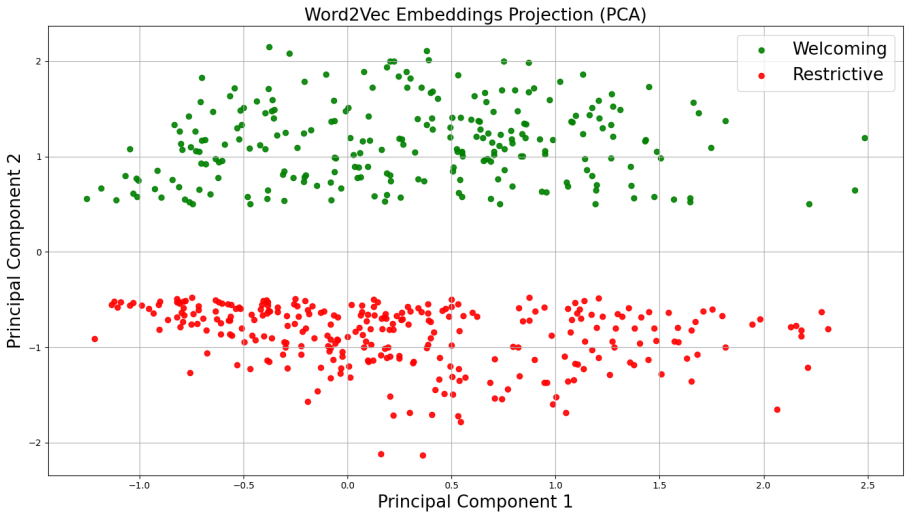


Figure 3: PCA projection of the refined dictionaries after y-coordinate threshold cleaning (restrictive = red, welcoming = green), showing clearer separation between the two stances.

not capture. Second, some dictionary words carried purely administrative or legal meaning (e.g., *Familiennachzug*, *Asylrecht*, *Ausländer*) or no clear stance at all (e.g., *abschieben*, *Abschiebung*, *Zurückweisung*, *Migrationswelle*, *Migrationsstrom*), adding noise unrelated to the speaker's actual position. We removed all such entries, retrained the model, and recomputed the scores.

This cleaning was not done to force parties into a preset ranking: after this step, Bündnis 90/Die Grünen's average sentiment became less negative, while Die Linke's stayed negative (Figure 4). This shows the correction came from real misclassifications found by close reading, not from adjusting until a target ranking appeared. We revisit this in the robustness check (Section 4.5).

This gives a final welcoming dictionary of 233 words, covering skilled labor, the economy, the labor market, human rights, humanitarian aid, protection, integration, education, language learning, citizenship, social cohesion, and diversity and inclusion. The final restrictive dictionary has 177 words, covering border security, state control, repatriation, deportation, legal enforcement, capacity limits, labor market protection, and concerns about social integration [2].

[2] To conserve space in the main document, the complete lists of welcoming and restrictive dictionary terms are provided in the replication code and supplementary data folder accompanying this thesis.

Table 4: Evolution of the welcoming and restrictive vocabularies

Stage	Welcoming	Restrictive	Total
Initial seeds	655	872	1527
Word2Vec expansion	755	968	1724
PCA refinement	241	288	529
Final vocabulary	233	177	410

3.3 Sentiment Index Construction

We build an immigration sentiment index from the reduced corpus. To capture semantic similarity in natural language, we need a method that turns sentences, paragraphs, and whole documents into numeric vectors while keeping their meaning. Since our corpus has thousands of speeches, Doc2Vec is a good fit: it is an unsupervised, neural-network method that learns vector representations of documents in a high-dimensional space (Le and Mikolov 2014).

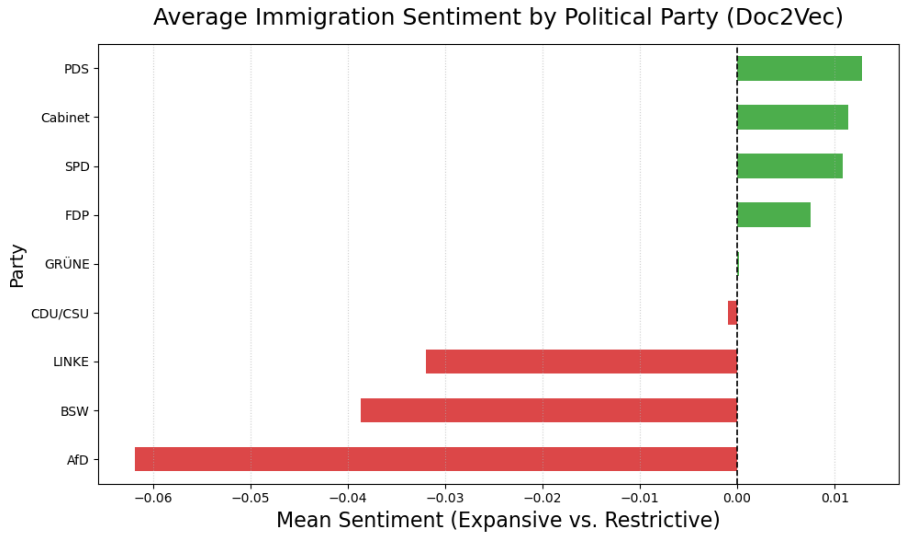


Figure 4: Average net immigration sentiment by political party (Doc2Vec measure). Negative values indicate that a party's speeches are, on average, closer to the restrictive than to the welcoming reference vector, whether the restrictive vocabulary is used defensively or critically.

Instead of simply counting dictionary words, we embed both speeches and dictionary terms in one shared vector space, and define each speech's stance as its cosine similarity to two reference vectors, one per dictionary. We use a rolling-window Doc2Vec procedure to avoid look-ahead bias: speeches in a given quarter are always scored using a model trained only on earlier data.

For each quarter t, we train a Paragraph Vector - Distributed Bag of Words (PV-DBOW) Doc2Vec model (`dm=0`, with word vectors trained jointly via `dbow_words=1`), vector size 100, context window 5, minimum word count 1, initial learning rate 0.025, and 50 training epochs, using only speeches from the preceding 40 quarters (ten years), $[t-40, t)$. This follows the rolling-window design of Latifi et al. (2024) for fiscal sentiment, adapted here for immigration. Figure 5 shows the procedure.

Using this model, we build two reference vectors, one for each dictionary, by averaging the embeddings of all dictionary terms present in that window's vocabulary. For multi-word entries (up to five words), we sum the word embeddings, and an entry is used only if all of its words are in the vocabulary. We then infer document vectors for all speeches in the following, held-out quarter t, and compute their cosine similarity to each reference vector, giving a $similarity_{welcoming}$ and a $similarity_{restrictive}$ score per speech.

The first training window covers Q1 1980 to Q4 1989, since immigration-related discourse becomes more consistent from 1980 onward. With the ten-year window requirement, this gives out-of-sample sentiment scores from Q1 1990 to Q2 2025.

The sentiment score of speech i is the difference between its two similarities:

$$\text{sentiment}_i = \text{similarity}_{\text{welcoming},i} - \text{similarity}_{\text{restrictive},i}. \tag{3.1}$$

A positive score means a speech is closer to the welcoming reference vector; a negative score means the opposite.

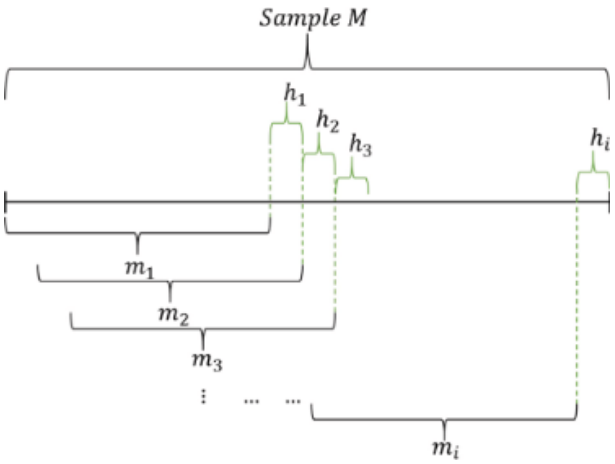


Figure 5: Rolling-window estimation for the Doc2Vec model. The corpus is split into fixed training windows (m) and following inference windows (h), shifted forward by one period each time. Adapted from Latifi et al. (2024).

Since the macroeconomic variables in Section 4.3 are observed at a monthly frequency, we aggregate the speech-level scores to months. Each speech is assigned to the calendar month in which it was delivered. For speaker group $g \in \{\text{Government}, \text{Opposition}\}$ and month m, the sentiment index is the unweighted mean of the scores of all immigration-related speeches given by members of that group:

$$S_{g,m} = \frac{1}{N_{g,m}} \sum_{i \in \mathcal{I}_{g,m}} \text{sentiment}_i, \tag{3.2}$$

where $\mathcal{I}_{g,m}$ is the set of speeches by group g in month m and $N_{g,m}$ their number. Every speech receives the same weight, regardless of its length or the speaker's role. A speech is assigned to the government group if the speaker's party belonged to the governing coalition on the date of the speech, and to the opposition otherwise. The overall parliamentary index is computed in the same way over all speeches.

Months without any immigration-related speech occur mainly during the parliamentary summer recess in July and August. Of the 426 months between January 1990 and June 2025, 46 contain no scored speech at all. The group-specific series can have additional empty months, when only government or only opposition members spoke on immigration. Because the VAR requires a continuous monthly series, we fill each empty month by linear interpolation between the two adjacent observed months. Gaps at the beginning or end of a series, where no neighbour exists on one side, are filled with the nearest observed value. The procedure is applied to the government, opposition, and overall series separately.

3.4 Validation of the Sentiment Measure

We check the plausibility of our sentiment measure by comparing it to well-known shifts in German immigration politics. A close match between the estimated series and known historical events is qualitative evidence that the measure captures real variation in political discourse, not just noise.

Overall sentiment shows two broad phases: mostly positive from 1990 to 2014, then clearly negative from 2014 onward (Figure 6). Government and opposition sentiment move together for most of the period, diverging mainly around 2006–2007 and 2018–2022 (Figure 9).

A clear dip appears between 1992 and 1994, matching the more restrictive rhetoric

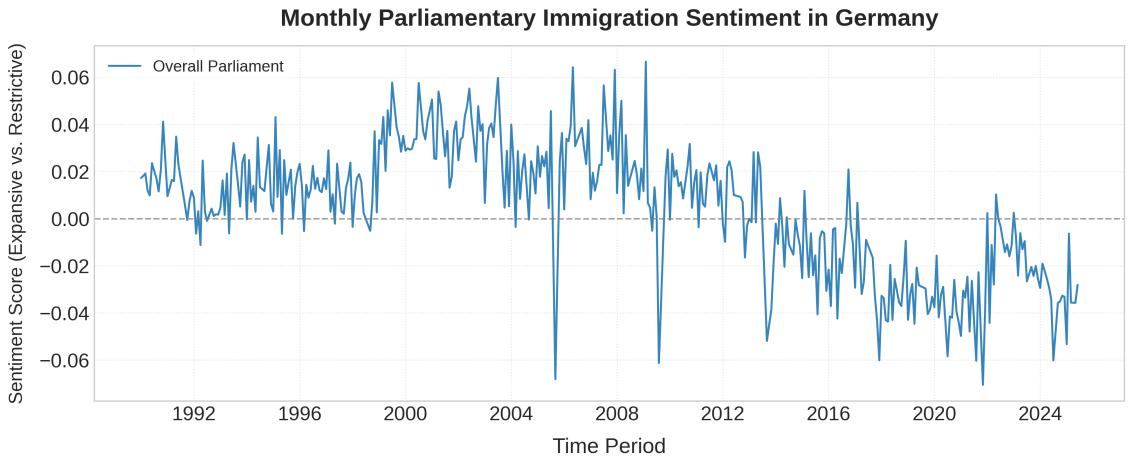


Figure 6: Monthly trend of overall parliamentary immigration sentiment in Germany. Positive scores show a net-expansive vocabulary; negative scores show more restrictive terminology (whether defensive or critical).

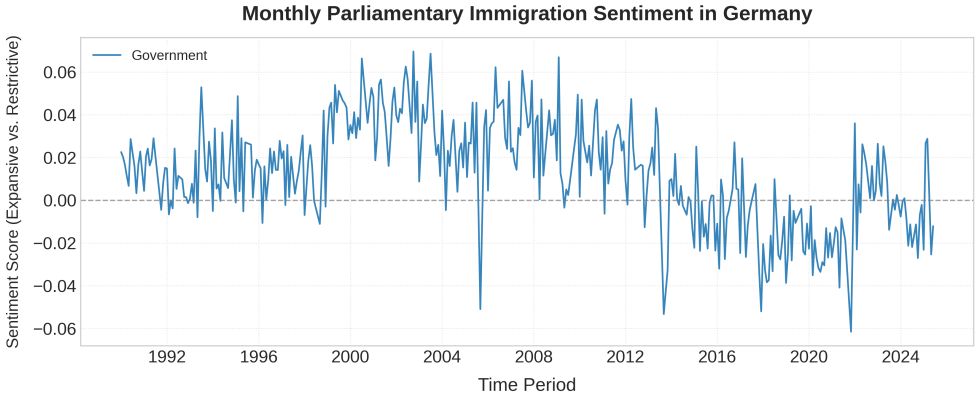


Figure 7: Monthly trend of government parliamentary immigration sentiment in Germany.

of that time aimed at reducing asylum applications (Borkert and Bosswick 2007). Sentiment turns more positive around 2000, matching the introduction of birthright citizenship (jus soli) and the Süssmuth Commission's work (Borkert and Bosswick 2007). Between 2014 and 2021, government sentiment swings more sharply (Figure 7), while opposition sentiment trends more negative (Figure 8) — patterns credibly linked to the Syrian refugee crisis (Trauner and Turton 2017), the COVID-19 pandemic, and the 2021 federal election, where immigration was a major campaign topic and public debate turned more restrictive. The positive peak in overall sentiment in 2022 (Figure 6) realistically reflect solidarity with Ukrainian refugees (Brücker 2022).

Party-level results show large differences in immigration-related discourse. The AfD shows mostly negative sentiment throughout its observed period, while parties tradi-

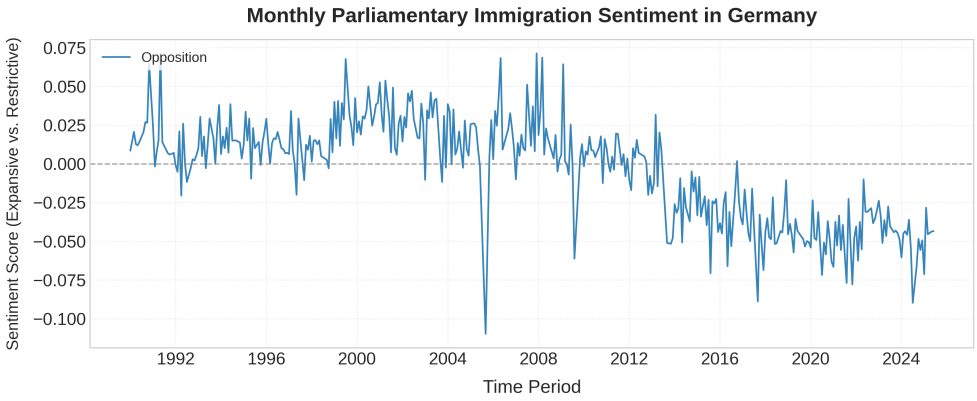


Figure 8: Monthly trend of opposition parliamentary immigration sentiment in Germany.

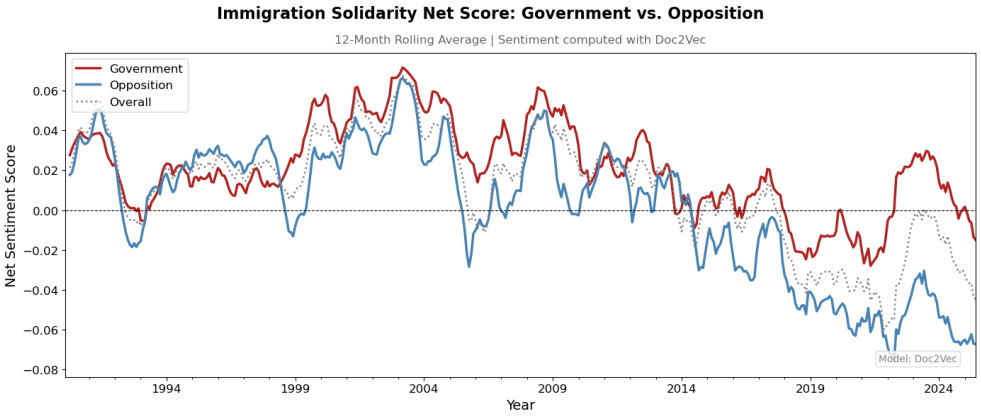


Figure 9: Monthly trend of immigration sentiment for the overall parliament, government, and opposition in Germany.

tionally seen as pro-immigration show higher average scores, though variation within each party remains substantial (Figure 34).

4 Estimating the macroeconomic effects

We now investigate the macroeconomic effects of immigration sentiment as reflected in the textual data. For that purpose, we augment the baseline VAR model from the study (Maffei-Faccioli and Vella 2021) by our new sentiment series.

4.1 VAR Model

This study employs a Bayesian vector autoregressive (BVAR) framework to analyse the dynamic relationships among the selected macroeconomic and political variables. The model is estimated over different sample periods in order to examine the robustness of the results and to distinguish between the pre-pandemic and extended sample periods. Specifically, the analysis considers the periods 2006–2019 (the period considered in the original study, because monthly immigration data are available only since 2006), 2007–2019, and 2007–2025 (we extended and reconstructed the dataset, beginning in 2007, because the employment rate is available on the website of the Bundesagentur für Arbeit only since May 2007).

For the pre-pandemic samples, namely 2006–2019 and 2007–2019, we estimate a standard reduced-form VAR model with p lags and n endogenous variables:

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{A}_1\mathbf{y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{y}_{t-p} + \mathbf{u}_t, \tag{4.1}$$

where $\mathbf{y}_t$ denotes an $n \times 1$ vector of endogenous variables, $\boldsymbol{\mu}$ is an $n \times 1$ vector of intercept terms, and $\mathbf{A}_i$, for $i = 1, \ldots, p$, are $n \times n$ autoregressive coefficient matrices. The reduced-form disturbance vector $\mathbf{u}_t$ is assumed to follow a multivariate normal distribution,

$$\mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{4.2}$$

where $\boldsymbol{\Sigma}$ is a positive-definite $n \times n$ variance–covariance matrix satisfying

$$E(\mathbf{u}_t\mathbf{u}_t') = \boldsymbol{\Sigma}. \tag{4.3}$$

The reduced-form disturbances are assumed to be serially uncorrelated at all non-zero leads and lags.

For the extended sample covering the period 2007–2025, the baseline VAR specification is augmented by an exogenous COVID-19 dummy variable in order to account for the exceptional economic and social disturbances associated with the pandemic. The extended model is therefore specified as

$$\mathbf{y}_t = \boldsymbol{\mu} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{B}\mathbf{x}_t + \mathbf{u}_t, \tag{4.4}$$

where $\mathbf{x}_t$ denotes the vector of exogenous variables and $\mathbf{B}$ contains the corresponding coefficients. In the present application, $\mathbf{x}_t$ consists of a single COVID-19 dummy variable, such that

$$COVID_t = \begin{cases} 1, & \text{for observations between March 2020 and October 2021,} \\ 0, & \text{otherwise.} \end{cases} \tag{4.5}$$

The COVID-19 dummy is included only in the estimation based on the 2007–2025 sample. It is treated as an exogenous control variable and does not form part of the endogenous VAR system. Consequently, it does not enter the autoregressive dynamics represented by the coefficient matrices $\mathbf{A}_1, \ldots, \mathbf{A}_p$. The VAR model can be written in compact matrix form as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\Pi} + \mathbf{U}, \tag{4.6}$$

where

$$\mathbf{Y} = \left(\mathbf{y}_{p+1}, \ldots, \mathbf{y}_T\right)' \tag{4.7}$$

is a $(T - p) \times n$ matrix containing the observations of the endogenous variables.

For the baseline VAR specification, the regressor matrix is given by

$$\mathbf{X} = \left(\mathbf{1}, \mathbf{Y}_{-1}, \ldots, \mathbf{Y}_{-p}\right), \tag{4.8}$$

where $\mathbf{1}$ is a $(T - p) \times 1$ vector of ones and

$$\mathbf{Y}_{-j} = \left(\mathbf{y}_{p+1-j}, \ldots, \mathbf{y}_{T-j}\right)' \tag{4.9}$$

is a $(T - p) \times n$ matrix containing the jth lag of the endogenous variables. Thus, in the

baseline specification, $\mathbf{X}$ has dimension

$$(T - p) \times (1 + np). \tag{4.10}$$

For the extended 2007–2025 specification, the exogenous COVID-19 dummy is included in the regressor matrix:

$$\mathbf{X} = \left(\mathbf{1}, \mathbf{X}_{COVID}, \mathbf{Y}_{-1}, \ldots, \mathbf{Y}_{-p}\right). \tag{4.11}$$

Since only one exogenous variable is included, the corresponding regressor matrix has dimension

$$(T - p) \times (2 + np). \tag{4.12}$$

More generally, with k exogenous variables, the dimension of the regressor matrix is $(T - p) \times (1 + k + np)$.

The coefficient matrix for the baseline specification is defined as

$$\mathbf{\Pi} = \left(\boldsymbol{\mu}, \mathbf{A}_1, \ldots, \mathbf{A}_p\right)', \tag{4.13}$$

whereas, for the extended specification including the COVID-19 dummy, it is given by

$$\mathbf{\Pi} = \left(\boldsymbol{\mu}, \mathbf{B}, \mathbf{A}_1, \ldots, \mathbf{A}_p\right)'. \tag{4.14}$$

The error matrix is defined as

$$\mathbf{U} = \left(\mathbf{u}_{p+1}, \ldots, \mathbf{u}_T\right)', \tag{4.15}$$

and has dimension $(T - p) \times n$.

4.1.1 Bayesian Estimation

Both the baseline VAR models and the extended specification including the COVID-19 dummy are estimated within a Bayesian framework.

Lag length was assessed using the Akaike Information Criterion (AIC), Hannan–Quinn Information Criterion (HQ), Schwarz Information Criterion (SC), and Final Prediction Error (FPE). As shown in Table 5, the criteria disagree substantially: AIC

and FPE favor long lag structures, particularly in the 2007–2019 and 2007–2014 subsamples, while SC consistently favors one or two lags across all samples. We follow the Schwarz Criterion throughout, both for its parsimony and to guard against overfitting given the limited sample size.

Table 5: Lag-order selection for the VAR in levels

Sample	Sentiment	AIC	HQ	SC	FPE
2007–2025	Government	3	3	3	3
2007–2025	Opposition	3	3	2	3
2007–2025	Overall	3	3	3	3
2007–2019	Government	5	3	2	5
2007–2019	Opposition	5	3	2	5
2007–2019	Overall	5	3	2	5
2007–2014	Government	4	2	1	4
2007–2014	Opposition	4	2	2	4
2007–2014	Overall	4	2	1	4

The objective is to generate posterior draws (one thousand) of the coefficient matrix $\mathbf{\Pi}$ and the covariance matrix $\mathbf{\Sigma}$ from their joint posterior distribution. According to Bayes' theorem, the joint posterior distribution is proportional to the likelihood of the observed data multiplied by the prior distributions:

$$p(\mathbf{\Pi}, \mathbf{\Sigma} \mid \mathbf{Y}, \mathbf{X}) \propto p(\mathbf{Y} \mid \mathbf{X}, \mathbf{\Pi}, \mathbf{\Sigma}) p(\mathbf{\Pi} \mid \mathbf{\Sigma}) p(\mathbf{\Sigma}). \tag{4.16}$$

Following a non-informative Bayesian approach, a flat prior is assigned to the coefficient matrix,

$$p(\mathbf{\Pi} \mid \mathbf{\Sigma}) \propto 1, \tag{4.17}$$

while the covariance matrix is assigned the Jeffreys prior,

$$p(\mathbf{\Sigma}) \propto |\mathbf{\Sigma}|^{-(n+1)/2}. \tag{4.18}$$

Posterior simulation is carried out by drawing from the corresponding conditional posterior distributions of $\mathbf{\Pi}$ and $\mathbf{\Sigma}$. By repeating this procedure thousands of times, we obtain posterior draws from the joint posterior distribution. A detailed derivation of the posterior distributions and a description of the Bayesian estimation procedure can be found in Maffei-Faccioli and Vella (2021).

4.1.2 Structural Identification

The reduced-form disturbances $\mathbf{u}_t$ are generally contemporaneously correlated. Therefore, a structural identification scheme is required in order to interpret the innovations as economically meaningful shocks. The relationship between the reduced-form disturbances and the structural disturbances is given by

$$\mathbf{u}_t = \mathbf{S}\boldsymbol{\varepsilon}_t, \tag{4.19}$$

where $\boldsymbol{\varepsilon}_t$ is an $n \times 1$ vector of structural disturbances satisfying

$$\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \tag{4.20}$$

Consequently, the covariance matrix of the reduced-form disturbances can be written as

$$\boldsymbol{\Sigma} = \mathbf{S}\mathbf{S}'. \tag{4.21}$$

Structural shocks are identified using a recursive Cholesky decomposition of $\boldsymbol{\Sigma}$, where $\mathbf{S}$ is the lower triangular matrix satisfying

$$\mathbf{S}\mathbf{S}' = \boldsymbol{\Sigma}. \tag{4.22}$$

The Cholesky decomposition is applied to the endogenous variables only. Accordingly, the exogenous COVID-19 dummy included in the extended 2007–2025 specification does not constitute an additional structural shock and does not affect the dimensionality or ordering of the structural innovation vector.

4.2 Data

The baseline specification has five endogenous variables: net immigration rate $NetImmgr_t$ [3]; log Business Expectations Index [4] $\log(BExp_t)$; log Consumer Confidence Index [5]

[3] The net migration rate (expressed per 1,000 inhabitants) is constructed using data from the German Federal Statistical Office (Statistisches Bundesamt (Destatis) 2026b; Statistisches Bundesamt (Destatis) 2026a). Because official net migration balance data from Destatis are consistently available from 2008 onward (Table 12711-0011), the pre-2008 historical observations (covering 2006 and 2007) are extended using the underlying monthly administrative migration series from Maffei-Faccioli and Vella (2021).

[4] Data regarding the business climate index are sourced from the Ifo Institute (Ifo Institute 2026).

[5] Data on consumer confidence are obtained from the Organisation for Economic Co-operation and Development (OECD 2026).

$\log(CConf_t)$; log Industrial Production Index [6] $\log(IndPr_t)$; and registered unemployment rate [7] $UnemplR_t$:

$$\mathbf{y}_t = \begin{bmatrix} NetImmgr_t \\ \log(BExp_t) \\ \log(CConf_t) \\ \log(IndPr_t) \\ UnemplR_t \end{bmatrix}. \tag{4.23}$$

So the baseline VAR has $n = 5$ endogenous variables. In the alternative specifications, we add one immigration sentiment indicator at a time, ordered last, giving six endogenous variables:

$$\mathbf{y}_t^{(s)} = \begin{bmatrix} NetImmgr_t \\ \log(BExp_t) \\ \log(CConf_t) \\ \log(IndPr_t) \\ UnemplR_t \\ S_{it} \end{bmatrix}, \qquad i \in \{Government, Opposition\}, \tag{4.24}$$

where S_{it} is immigration sentiment for speaker group i.

We follow Maffei-Faccioli and Vella (2021) in choosing the macroeconomic variables. Consumer confidence and business expectations are forward-looking indicators, included because migration decisions may depend on expected, not just current, employment and income conditions; adding forward-looking variables also helps limit the risk of non-fundamentalness (Forni and Gambetti 2014).

The Consumer Confidence Index comes from household assessments of their financial situation, the general economy, employment prospects, and savings, sourced from the Organisation for Economic Co-operation and Development (OECD). The Business Expectations Index comes from monthly ifo Institute surveys of firms in manufacturing, services, trade, and construction, on their expectations for the next six months. Industrial production captures overall economic activity, and registered unemployment lets us directly examine the link between labour-market conditions and immigration.

[6] Data on industrial production are obtained from the Deutsche Bundesbank statistical time-series database (Deutsche Bundesbank 2026).

[7] The unemployment rate is the share of registered unemployed in the labor force, where the labor force is employed German residents (Destatis) plus registered unemployed (Bundesagentur für Arbeit).

All log series are multiplied by 100, and the data are monthly, so that coefficients and impulse responses can be read directly as percentage changes. Our analysis reconstructs and extends the dataset from Maffei-Faccioli and Vella (2021).

We estimate three VAR models over three sample periods. First, 2006–2019, matching the original study data. Second, 2007–2019 on our reconstructed data, motivated by a large discrepancy we found between the original study's consumer confidence series and the series from its stated source (Figure 10); this lets us check whether results are sensitive to our reconstructed series while staying close to the original sample. Third, the extended 2007–2025 period, adding the COVID-19 pandemic and its aftermath, with a binary COVID-19 dummy as an exogenous control (Eq. 4.5) to keep pandemic-driven disruption from being wrongly attributed to the endogenous relationships between immigration, the economy, and political sentiment. This dummy appears only in the 2007–2025 specification and is excluded from both the endogenous vector and the structural shocks.

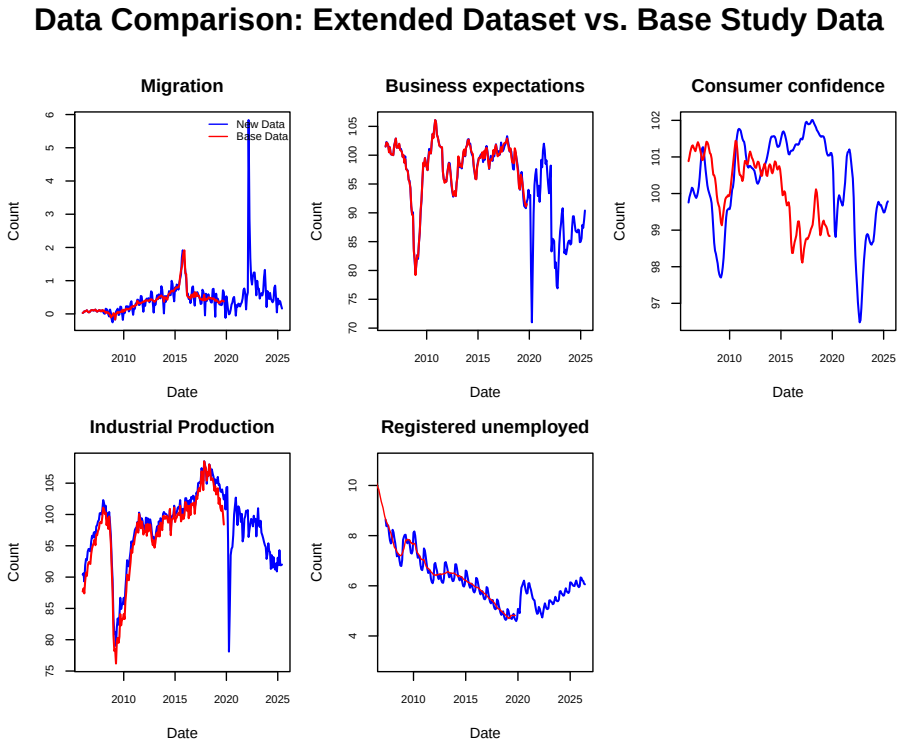


Figure 10: Data Comparison: Extended Dataset vs. Base Study Data. Five time-series charts (2010–2025) comparing New Data (blue) and Base Data (red) for Migration, Business expectations, Consumer confidence, Industrial Production, and Registered unemployed.

4.3 Identification

We identify immigration and sentiment shocks with a recursive (Cholesky) ordering, following (Maffei-Faccioli and Vella 2021) for the migration block and extending it by placing our new Bundestag immigration sentiment measure last, as in Eq. 4.24. We report two specifications, one per sentiment measure ($i \in \{Government, Opposition\}$); only the sentiment series and the lags number changes across them, so the migration and macro block ordering stays fixed, and any difference in results can be attributed to the sentiment measure itself, not the identification scheme.

Arrivals are ordered first, assuming German economic conditions do not affect migration within the same month: as (Maffei-Faccioli and Vella 2021) argue, deciding to migrate, getting authorization, and arriving takes longer than a month, so a same-month economic shock is unlikely to drive contemporaneous migration.

We order $\mathrm{BExp}_t, \mathrm{CConf}_t, \mathrm{IndPr}_t, \mathrm{UnemplR}_t$, following (Maffei-Faccioli and Vella 2021): the forward-looking survey measures (expectations, confidence) come before the realized ("harder") indicators (production, unemployment). This internal ordering is not critical to our results, since both survey and realized variables sit between arrivals and sentiment in every specification; reordering within this block does not affect the sentiment ordering or the identification of the sentiment shock, our main object of interest.

Sentiment is ordered last because parliamentary immigration rhetoric is likely to respond contemporaneously to prevailing economic and social conditions, while its effects on migration and macroeconomic outcomes are expected to operate with a delay through public discourse and social amplification (Abdelatif and Andriantomanga 2026; Rakers 2023). The recursive identification therefore assumes that macroeconomic and migration variables may affect sentiment within the month, but sentiment does not affect them contemporaneously. This is an identifying assumption, which we assess through alternative recursive orderings in Section 4.5.

We estimate the VAR in levels, as our interest lies in short- to medium-run dynamics. KPSS tests reject level stationarity for all variables at the 1% level, except log industrial production (Tables 6 in the Appendix), and Johansen trace tests indicate at least two cointegrating relations in all samples and sentiment specifications (Tables 7 in the Appendix). Estimation in levels remains consistent in the presence of unit roots and cointegration Sims, Stock, and Watson 1990, whereas differencing would discard

the long-run information contained in the data. The impulse responses are therefore interpreted as dynamic associations rather than as structural causal effects.

4.4 Results

All impulse responses refer to a one-standard-deviation shock. Because log series are multiplied by 100, responses of business expectations, consumer confidence and industrial production are in percent, the response of the unemployment rate is in percentage points, and arrivals are in net migrants per 1,000 inhabitants. Since sentiment is ordered last, the shock is the part of sentiment that is not explained by past values of all variables or by same-month macroeconomic and migration conditions. It is best read as an unexpected shift toward a more welcoming (positive) or more restrictive (negative) vocabulary, not as a change in policy. We call a response credible when the 90% credible band excludes zero, and we mention responses that are supported only by the 68% band separately.

On the baseline data set, both sentiment shocks are followed by a temporary economic slowdown (Figures 11 and 12). After a government sentiment shock, business expectations fall by about 0.5% within five months, consumer confidence falls by about 0.06%, and industrial production falls by about 0.5% after roughly a year. The unemployment rate rises by about 0.03 percentage points, with a peak after about 15 months. The opposition shock produces a similar pattern, with a slightly larger fall in business expectations (about 0.6%) and a delayed decline in industrial production. In both models, the responses turn slightly positive after about two years, so the shocks have no lasting effect on the level of activity.

On the reconstructed data set, most of these responses lose credibility (Figures 13 and 14). The responses of business expectations, consumer confidence and industrial production keep their negative sign, but their credible bands largely include zero. The exception is the unemployment rate. It rises by about 0.07 percentage points within three months after both shocks and remains above zero over the following years. Two factors can explain the weaker results. The sample is shorter, and the reconstructed consumer confidence series differs markedly from the one used in the original study (Figure 10). The baseline results should therefore be read with some caution, since they partly depend on the earlier version of the data.

Over the full period, the responses become larger, and the two speaker groups di-

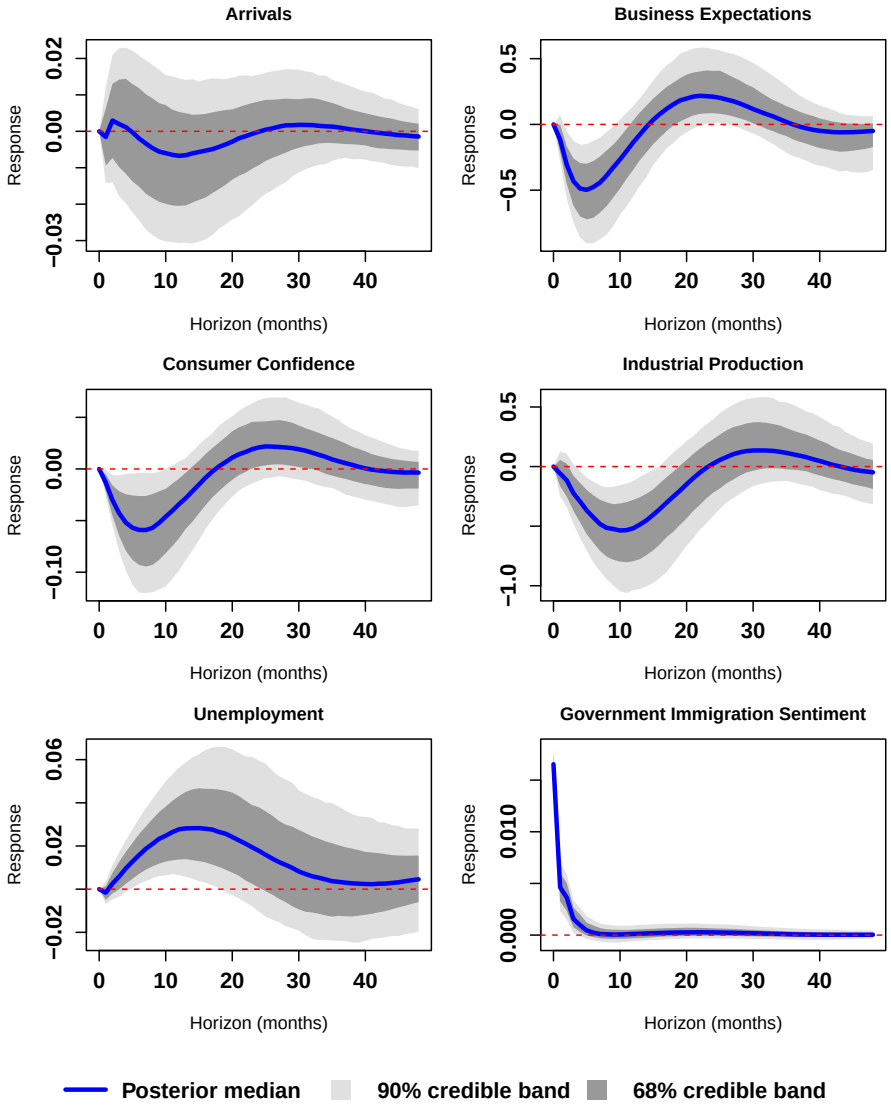


Figure 11: Impulse responses to a one-standard-deviation government immigration sentiment shock, baseline panel, 2006M1–2019M10 ($n = 166$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

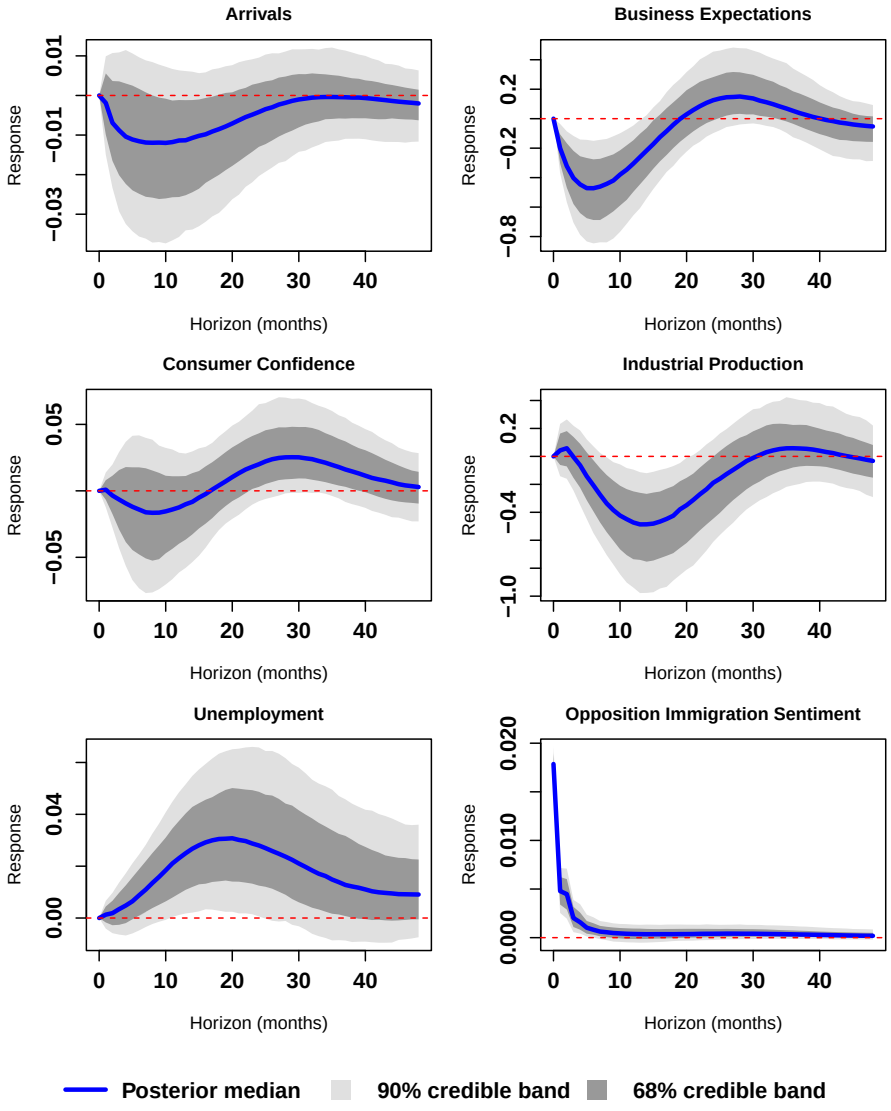


Figure 12: Impulse responses to a one-standard-deviation opposition immigration sentiment shock, baseline panel, 2006M1–2019M10 ($n = 166$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

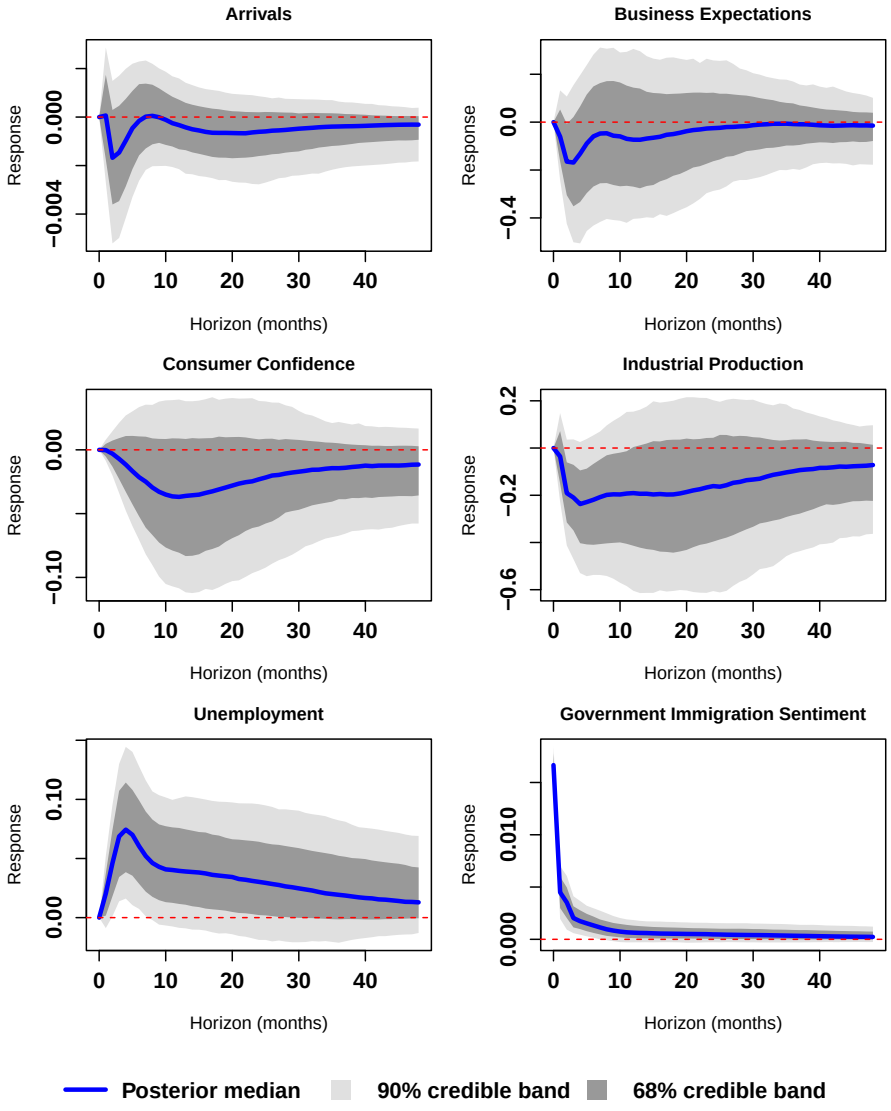


Figure 13: Impulse responses to a one-standard-deviation government immigration sentiment shock, new data panel, 2007M5–2019M10 ($n = 150$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

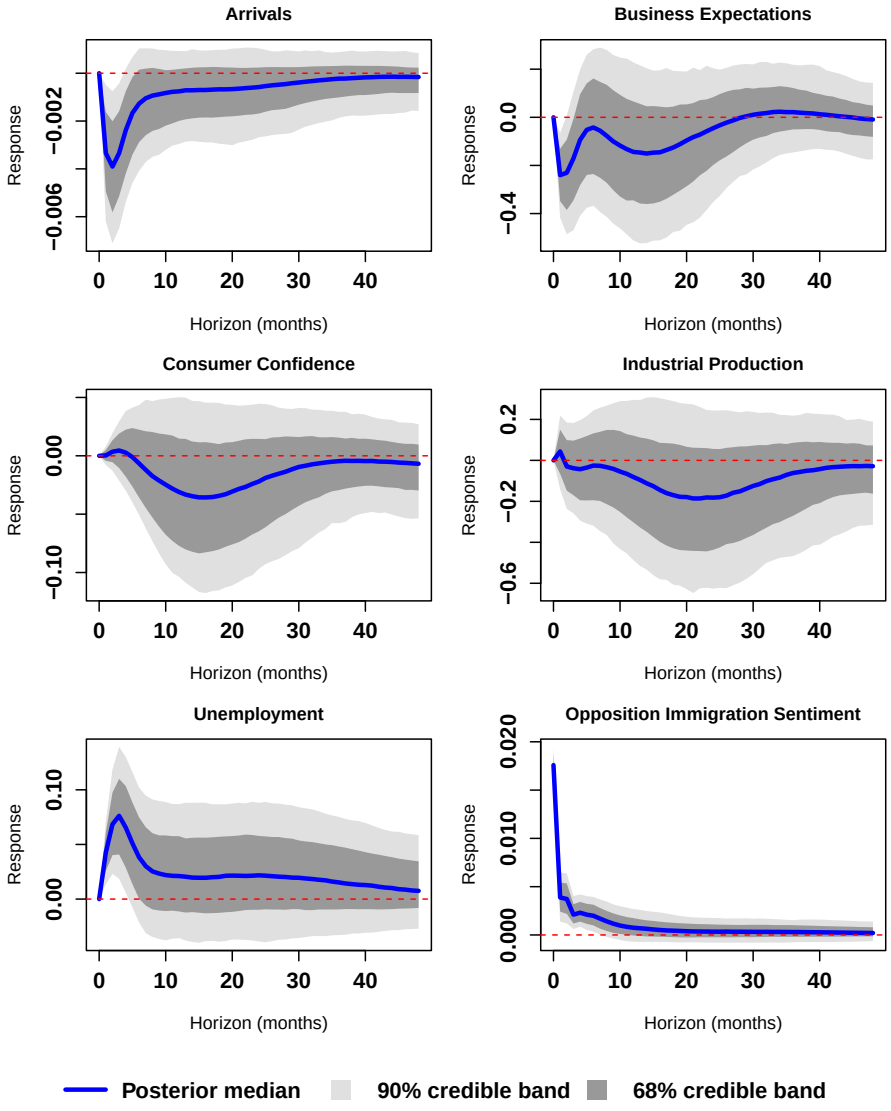


Figure 14: Impulse responses to a one-standard-deviation opposition immigration sentiment shock, new data panel, 2007M5–2019M10 ($n = 150$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

verge (Figures 15 and 16). After a government sentiment shock, business expectations fall by about 0.6% within three months, and consumer confidence and industrial production decline briefly. After an opposition sentiment shock, by contrast, business expectations rise by about 0.35% and consumer confidence by about 0.06%, both peaking after roughly ten months. Arrivals fall on impact and return to zero within about a year. In both models, the unemployment rate rises: by about 0.05 percentage points after a government shock and by about 0.08 percentage points after an opposition shock, with a peak after three to five months. This rise in unemployment is the most stable result across all samples and models.

The opposing signs for business expectations and consumer confidence are partly a result of how the two groups changed over the sample. Until 2017, the opposition consisted mainly of parties with welcoming positions on immigration. Since then, it has included the AfD, and from 2021 also the CDU/CSU. A positive opposition shock in the later years therefore often means that restrictive opposition rhetoric softened. Survey-based expectations may react more favourably to such a shift than to welcoming statements by a government that is responsible for managing large inflows. This interpretation is consistent with the instability of the opposition responses across samples. In 2006–2019, business expectations fell after an opposition shock, whereas in 2007–2025 they rose. Results for the opposition therefore depend on which parties form it.

That a shift toward more welcoming rhetoric is followed by higher unemployment is the most counterintuitive result, and several mechanisms are possible. The first is a composition effect in the unemployment statistics. Welcoming rhetoric is concentrated in periods of large refugee inflows, such as 2015–2016 and 2022. Refugees enter registered unemployment with a delay, once their asylum procedures are decided or, for refugees from Ukraine, after they moved into the basic income support system in mid-2022. Under this mechanism, sentiment does not affect the labour market directly. It signals inflows that appear in the unemployment statistics a few months later. The second mechanism is a mismatch between the skills of the unemployed and the needs of firms. Even when unemployment rises, political discourse may emphasise the need for skilled immigration in specific occupations and a more selective immigration policy. This would also be consistent with the temporary decline in arrivals after an opposition shock. The third possibility is anticipation. Politicians may react to labour-market

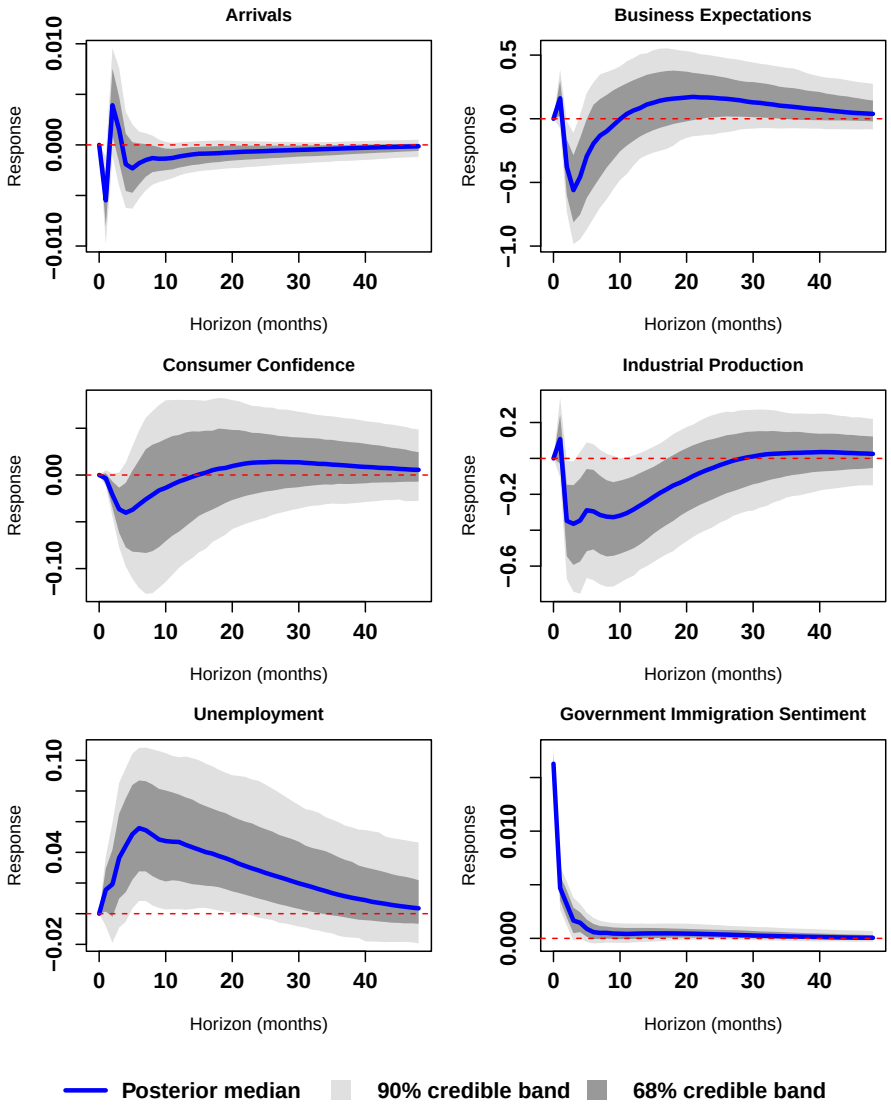


Figure 15: Impulse responses to a one-standard-deviation government immigration sentiment shock, new data panel, 2007M5–2025M6 ($n = 218$, $p = 3$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

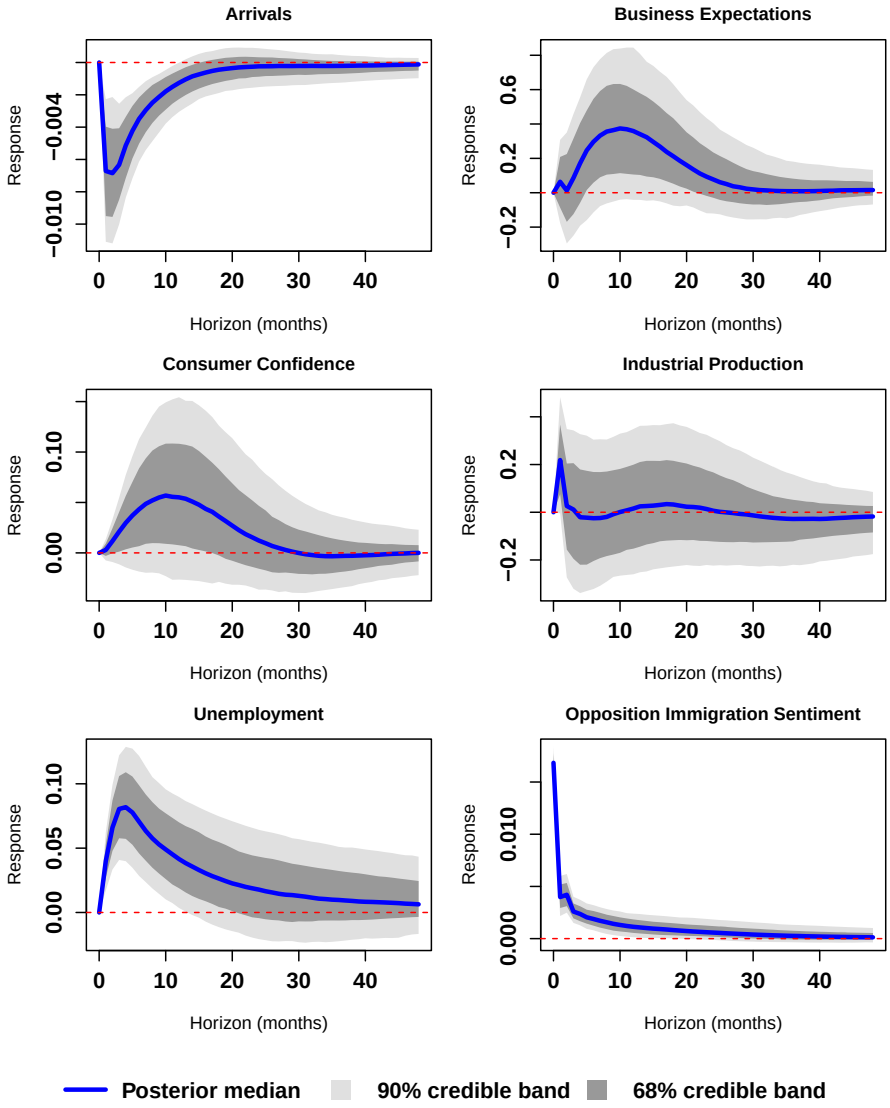


Figure 16: Impulse responses to a one-standard-deviation opposition immigration sentiment shock, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

information, such as announced layoffs or forecasts, before it appears in the monthly data. Distinguishing between these mechanisms would require unemployment rates for German and foreign nationals separately, which we leave for future work (Section 5).

The relationship also runs from the economy to sentiment. After a one-standard-deviation unemployment shock, government sentiment rises by about 0.002 and opposition sentiment by about 0.004, with a peak after three to four months (Figures 17 and 18). For the opposition, this corresponds to roughly a quarter of a standard deviation of the sentiment shock. The forecast error variance decomposition indicates that this channel is quantitatively relevant: after about one year, unemployment shocks account for roughly 20% of the forecast error variance of opposition sentiment (Figure 32). At first sight, this result contrasts with Abdelatif and Andriantomanga (2026), who find that higher economic uncertainty is followed by more restrictive rhetoric. The two findings need not contradict each other, however. Uncertainty and unemployment are different shocks, and our sample extends beyond 2018. In this later period, labour shortages persisted in many occupations, and skilled immigration became a prominent policy issue. The Skilled Immigration Act came into force in 2020 (Deutscher Bundestag 2019), and its reform followed in several steps between November 2023 and June 2024 (Deutscher Bundestag 2023a). Parliamentary debate in this period may therefore have linked weaker labour-market conditions with the need for targeted immigration rather than with restriction alone.

As a reference point, Figures 30 and 31 in the Appendix show the responses to a shock in arrivals itself. Because arrivals are ordered first in both specifications, this shock is identified in the same way in both models, which differ only in the sentiment series and the lag length. Two responses are common to both models. The unemployment rate falls by about 0.09 to 0.10 percentage points, with a trough after two to three months, and industrial production rises by about 0.35% within the first half year. Since Maffei-Faccioli and Vella (2021) report 68% bands, we compare at this level. Both responses are credible at 68% in both models, and their signs match the main findings of Maffei-Faccioli and Vella (2021) that migration shocks lower unemployment and expand economic activity, which supports the reconstructed data set. Under our stricter 90% criterion, the fall in unemployment is credible for about two and a half years in the opposition specification but only for the first three months in the government specification, and the rise in industrial production is credible only in the opposition

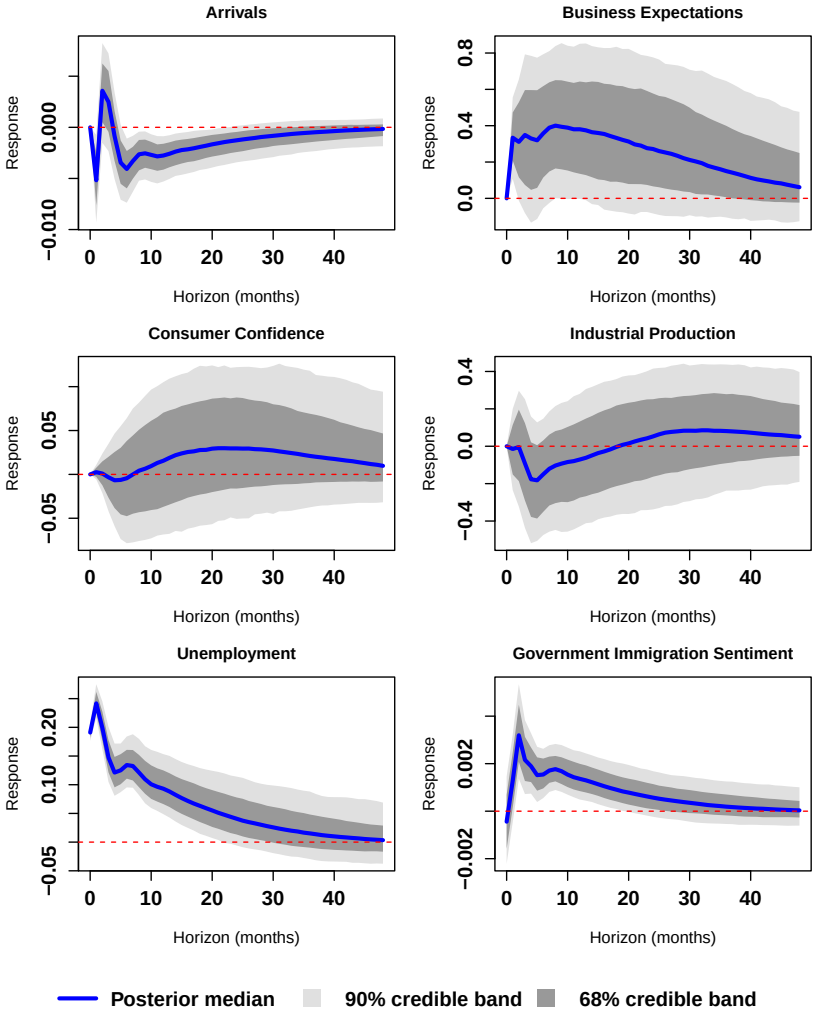


Figure 17: Impulse responses to a one-standard-deviation unemployment shock, new data panel, 2007M5–2025M6 ($n = 218$, $p = 3$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (variable ordering: arrivals, business expectations, consumer confidence, industrial production, unemployment, government immigration sentiment). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

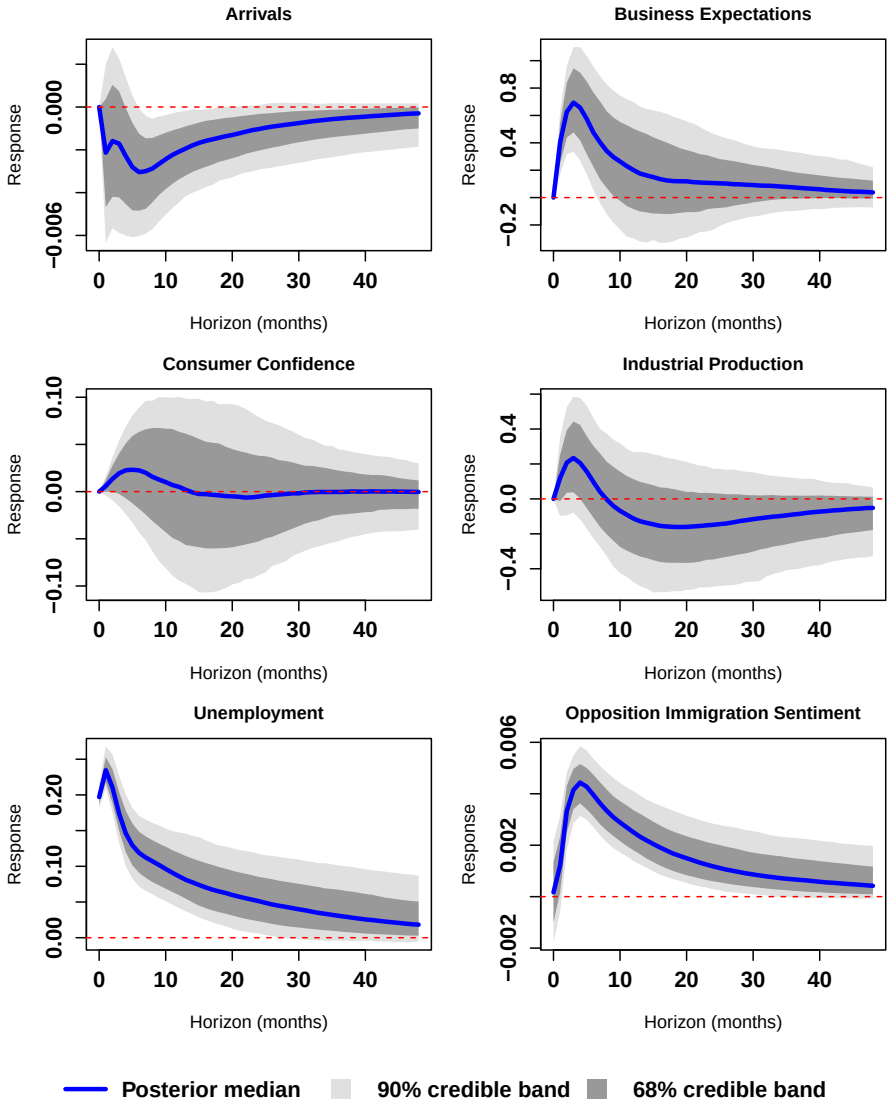


Figure 18: Impulse responses to a one-standard-deviation unemployment shock, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (variable ordering: arrivals, business expectations, consumer confidence, industrial production, unemployment, opposition immigration sentiment). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

specification (months 5 to 15). Compared with Maffei-Faccioli and Vella (2021), the responses in our sample also occur much faster, peaking within a few months rather than after one to two years.

Other responses differ from the original study and are not credible in our estimates. Business expectations fall on impact in both models and then turn briefly positive, but the positive phase is not credible even at the 68% level. By contrast, Maffei-Faccioli and Vella (2021) find an immediate increase. A plausible explanation is that our sample includes 2022, when the largest arrivals shock in the data coincided with the outbreak of the war in Ukraine and a sharp decline in business expectations. Since arrivals are ordered first, this coincidence enters the impact response. The response of consumer confidence is not credible in either specification, so we cannot confirm the negative response reported by Maffei-Faccioli and Vella (2021), and we do not interpret it further.

Finally, an arrivals shock is followed by a small shift toward more restrictive sentiment in both models. For opposition sentiment, this response is credible at the 90% level from about the third month onward; for government sentiment, only the 68% band excludes zero. This is in line with Abdelatif and Andriantomanga (2026), who find that migration-related developments are followed by more restrictive parliamentary rhetoric. Taken together, immigration sentiment and realised migration are associated with opposite movements in unemployment: the unemployment rate rises after a sentiment shock but falls after an arrivals shock. The sentiment shock therefore captures information that differs from the migration flow itself.

Three points qualify these results. First, the responses are dynamic associations, not causal effects (Section 4.3). Second, empty months in the sentiment series are filled by linear interpolation, which uses the value of the following month. This smooths the series and introduces a small amount of future information. Third, the government and opposition models for 2007–2025 use different lag lengths (three and two), as selected by the Schwarz criterion, so differences between them do not stem from the sentiment series alone. Section 4.5 examines whether the main findings survive a different sample period, an alternative ordering, and an independent sentiment measure.

4.5 Robustness

In this section, we examine the model further. First, we consider the pre-crisis period 2007–2014 to investigate whether the responses of the macroeconomic variables result from the refugee crisis rather than from immigration sentiment. Second, we revisit the identification by ordering immigration sentiment first in the model. Finally, we test the consistency of our results by recomputing immigration sentiment with the approach of Kostikova et al. (2024).

Figures 19 and 20 show that the responses of the endogenous variables to immigration sentiment are already notable in the period 2007–2014. Business expectations, consumer confidence and industrial production show distinctly credible responses: they are negative immediately after a government immigration sentiment shock, whereas their responses to an opposition shock are not credible, as the credible bands include zero. The unemployment rate still responds positively, but less credibly.

Ordering immigration sentiment first allows it to affect the other variables contemporaneously. The results remain qualitatively unchanged for the period 2007–2025 (Figures 21, 22), indicating that our previous interpretation does not depend on the identification assumption.

A separate concern, raised in Section 3.2, is that Doc2Vec struggles with negation. Manual inspection found several misclassified speeches, and even after the correction, average sentiment for some parties (CDU/CSU, Bündnis 90/Die Grünen, Die Linke) still looked As a check on the sentiment measure itself, we re-estimate immigration sentiment with a large language model (GPT-4.1-nano), adapting the approach of Kostikova et al. (2024). The Doc2Vec measure scores a speech by its lexical proximity to two reference vocabularies. The LLM, by contrast, classifies the stance a speaker takes toward migrants in context. It can therefore distinguish the endorsement of restrictive positions from their quotation or criticism, which is the main weakness identified in Section 3.2. Each migration-related excerpt is classified as solidarity, anti-solidarity, mixed, or none, and solidarity and anti-solidarity statements are additionally assigned to one subtype (Table 8; the full prompt is given in Appendix 9). For speaker group g (government or opposition) in month m, the net-solidarity score is

$$NS_{gm} = \frac{S_{gm} - A_{gm}}{N_{gm}}, \tag{4.25}$$

where S_{gm} and A_{gm} count the excerpts labelled solidarity and anti-solidarity, and N_{gm} is the total number of migration-related excerpts in that month. Because mixed and neutral excerpts enter the denominator, the score lies between -1 and 1 and reflects both the direction of stance and how often a clear stance is taken.

We place more weight on this measure than on the Doc2Vec measure for three reasons. First, Kostikova et al. (2024) show on Bundestag speeches that a large language model (GPT-4) reproduces human high-level (anti-)solidarity labels toward migrants with a macro F1 of 0.73, close to the human upper bound of 0.78. Solidarity and anti-solidarity are confused in only about 1% of cases. Our score in Eq. (4.25) relies only on this high-level distinction, which is the dimension where the Doc2Vec measure is weakest. Second, the party patterns produced by GPT-4.1-nano agree with the findings of Kostikova et al. (2024), which were validated against human annotations (see below). Third, a context-based classification does not depend on the composition of a fixed vocabulary, so the dictionary choices described in Section 3.2 do not affect it.

This evidence should nevertheless be read with caution. Kostikova et al. (2024) validate GPT-4, not the smaller GPT-4.1-nano used here, and they report clearly lower performance for smaller models. Our prompt also differs from theirs. It distinguishes economic-utility and economic-burden subtypes instead of a single exchange-based category, and it asks for a short justification instead of step-by-step reasoning. We therefore treat the LLM-based measure as the more valid indicator of stance, but not as a validated ground truth.

Figure 23 shows that the LLM-based measure ranks parties in line with their known positions on immigration. The AfD (-0.70) and the BSW (-0.43) are the only parties with negative net solidarity, while the SPD, Bündnis 90/Die Grünen and Die Linke have the highest scores. This differs from the Doc2Vec measure, which assigns Die Linke a clearly negative average (Figure 4). The difference supports our earlier suspicion that some discrepancies in Section 3.2 stem from negation and the critical use of restrictive vocabulary. The subtype shares show a similar pattern (Figures 24 and 25). Die Linke and Bündnis 90/Die Grünen most often express compassionate solidarity, whereas the FDP most often justifies immigration by its economic utility. This mirrors the party distribution reported by Kostikova et al. (2024), in which left-leaning parties emphasise compassionate solidarity, centrist and liberal parties emphasise economic contributions, and the AfD predominantly expresses anti-solidarity. Because even GPT-4

reaches only a fine-grained macro F1 of 0.42 in their evaluation, we interpret the subtype shares descriptively.

The impulse responses based on the LLM measure confirm the main pattern of Section 4.4. They also show more clearly that opposition sentiment carries more information than government sentiment. In the full sample 2007–2025, a one-standard-deviation opposition sentiment shock is followed by a rise in business expectations of about 1.2%, peaking after roughly seven months. Consumer confidence rises by about 0.2% with a similar timing, and the unemployment rate rises by about 0.10 percentage points, peaking after three to four months. Arrivals decline temporarily (Figure 29). The same signs already appear in 2007–2019, though with smaller magnitudes and wider bands (Figure 27). Together with the Doc2Vec results (Figure 16), this gives a robust core of findings that holds under both sentiment measures. After a shift toward a more welcoming opposition stance, unemployment, business expectations and consumer confidence rise, and arrivals fall temporarily. The response of industrial production differs between the two measures, so we do not interpret it further.

By contrast, government sentiment shocks are followed by weaker responses whose credible bands mostly include zero, in both 2007–2019 and 2007–2025 (Figures 26 and 28). One possible explanation is that government speakers are bound to the coalition's policy, so their stance changes less in response to new information. Opposition sentiment varies more strongly, partly because the composition of the opposition changes over the sample. Figure 23 shows that opposition net solidarity falls sharply after 2017, when the AfD entered the Bundestag, while government net solidarity remains positive. Opposition sentiment may therefore capture shifts in the broader political climate that government statements do not reflect.

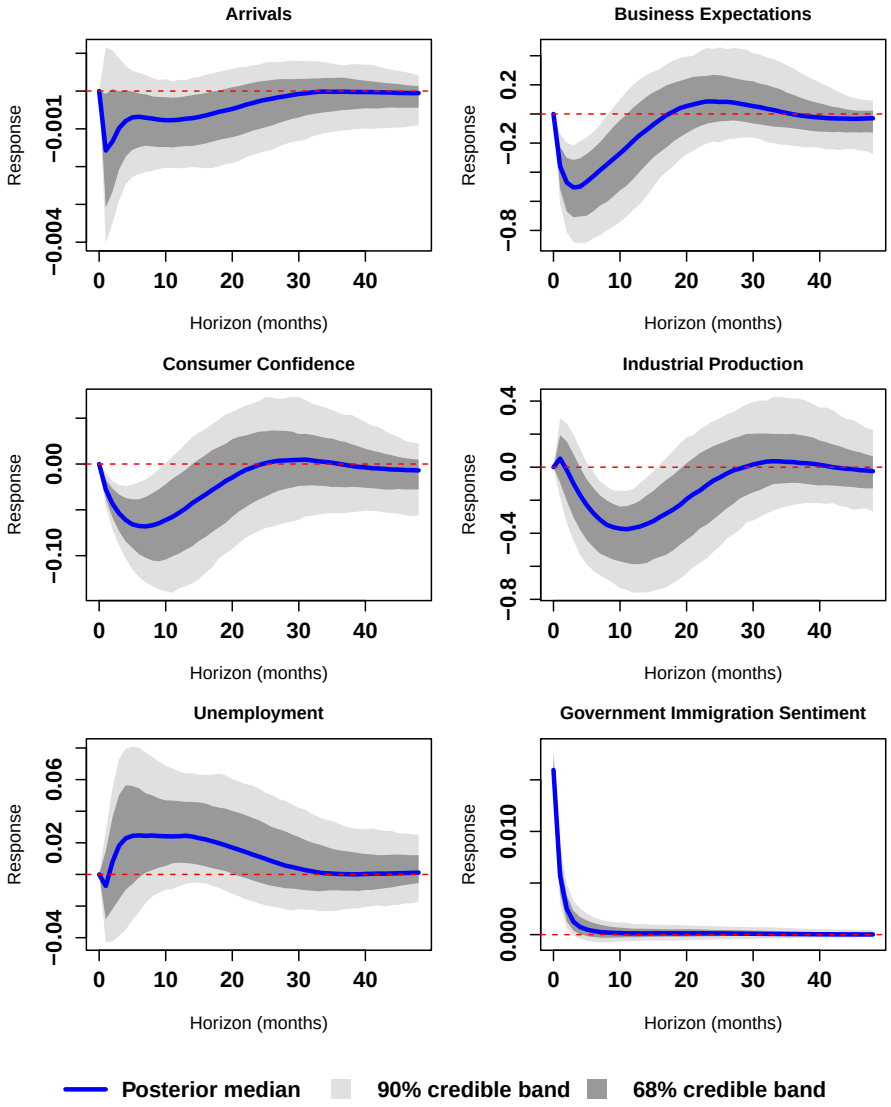


Figure 19: Impulse responses to a one-standard-deviation government immigration sentiment shock, new data panel, 2007M5–2014M12 ($n = 92$, $p = 1$ lag). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

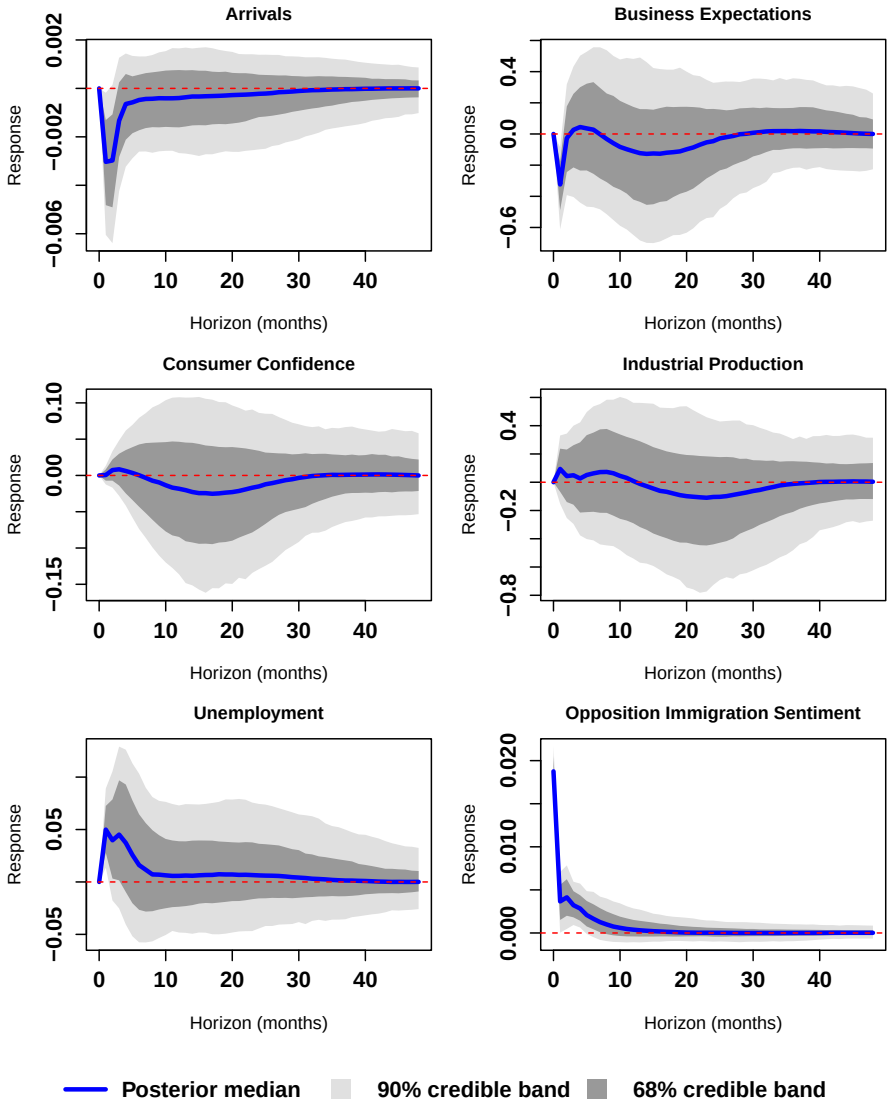


Figure 20: Impulse responses to a one-standard-deviation opposition immigration sentiment shock, new data panel, 2007M5–2014M12 ($n = 92$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

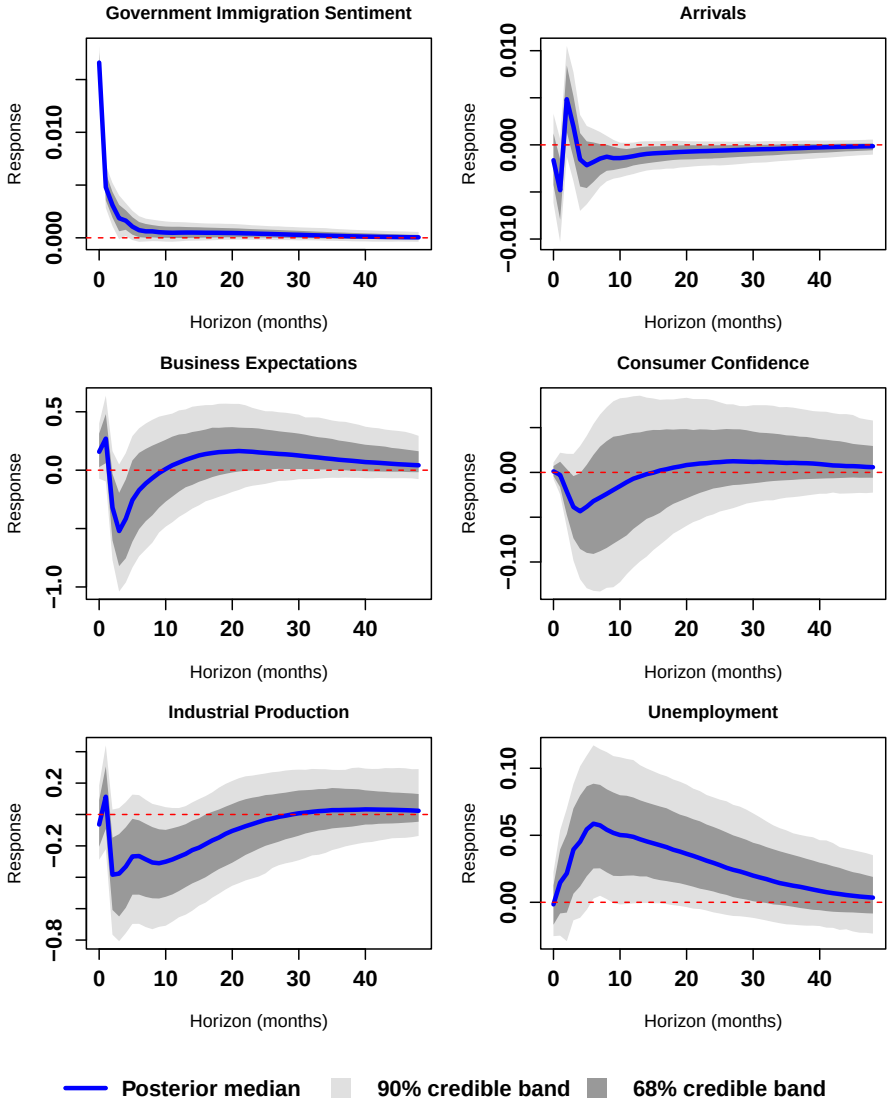


Figure 21: Robustness to the variable ordering: impulse responses to a one-standard-deviation government immigration sentiment shock with sentiment ordered first, new data panel, 2007M5–2025M6 ($n = 218$, $p = 3$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification. Lines show the posterior median; shaded areas show 68% and 90% credible bands.

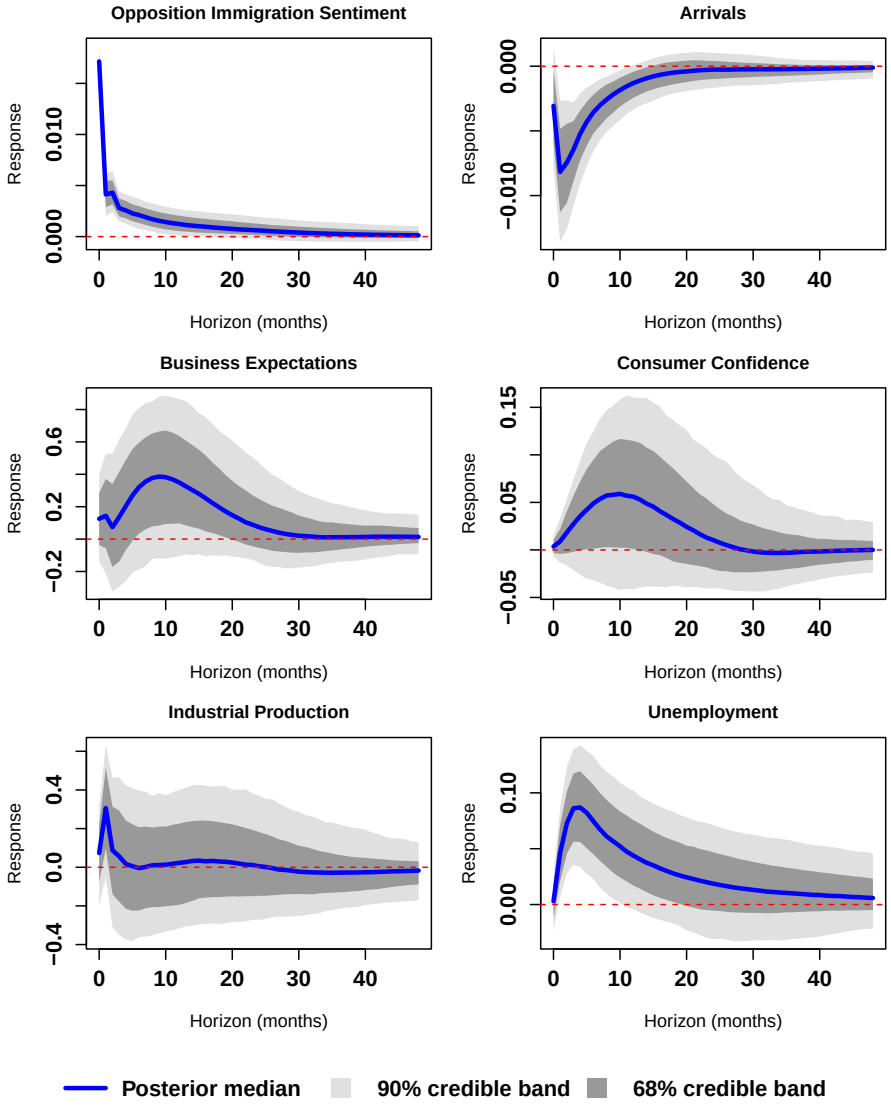


Figure 22: Robustness to the variable ordering: impulse responses to a one-standard-deviation opposition immigration sentiment shock with sentiment ordered first, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification. Lines show the posterior median; shaded areas show 68% and 90% credible bands.

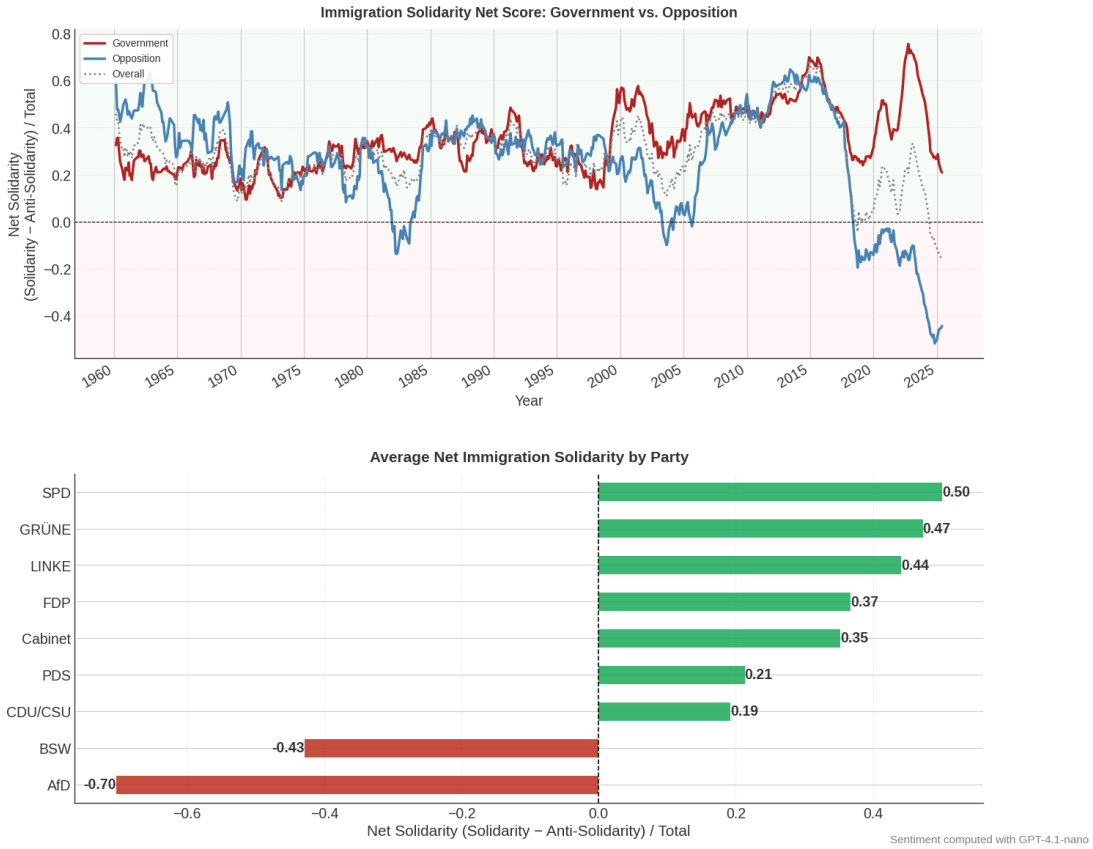


Figure 23: Net immigration solidarity over time (top) and average net immigration solidarity by party (bottom), GPT-4.1-nano.

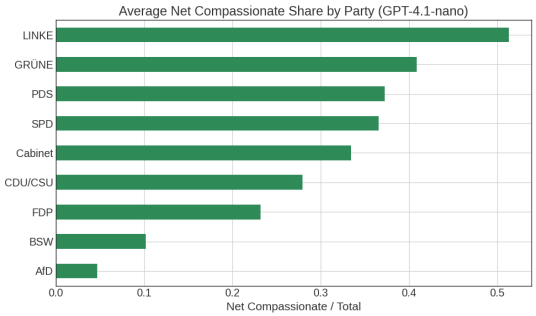


Figure 24: Average net compassionate solidarity by party (GPT-4.1-nano).

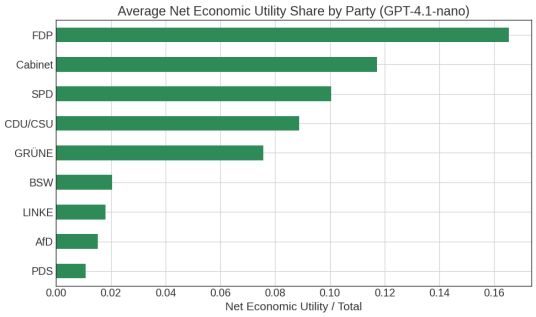


Figure 25: Average net economic-utility solidarity by party (GPT-4.1-nano).

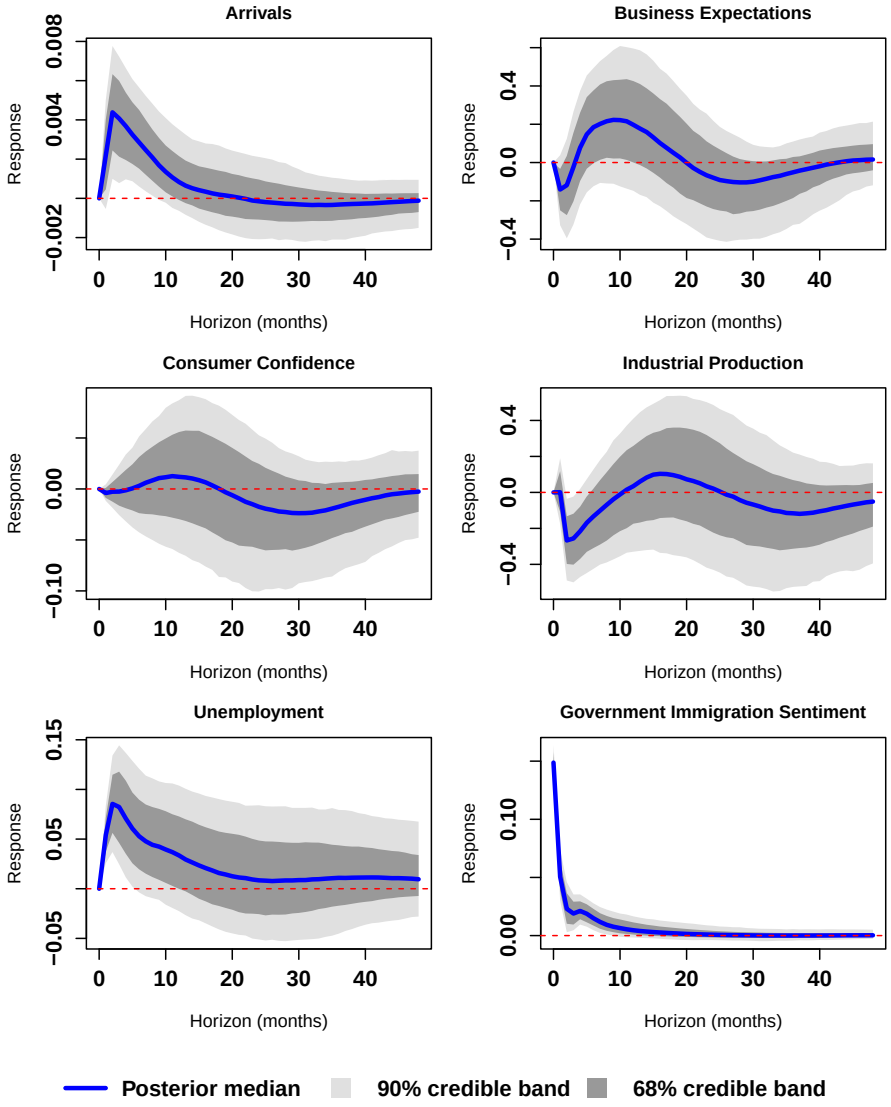


Figure 26: Robustness to the sentiment measure: impulse responses to a one-standard-deviation government immigration sentiment shock, with sentiment scored by GPT-4.1-nano instead of the doc2vec-based measure, new data panel, 2007M5–2019M10 ($n = 150$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

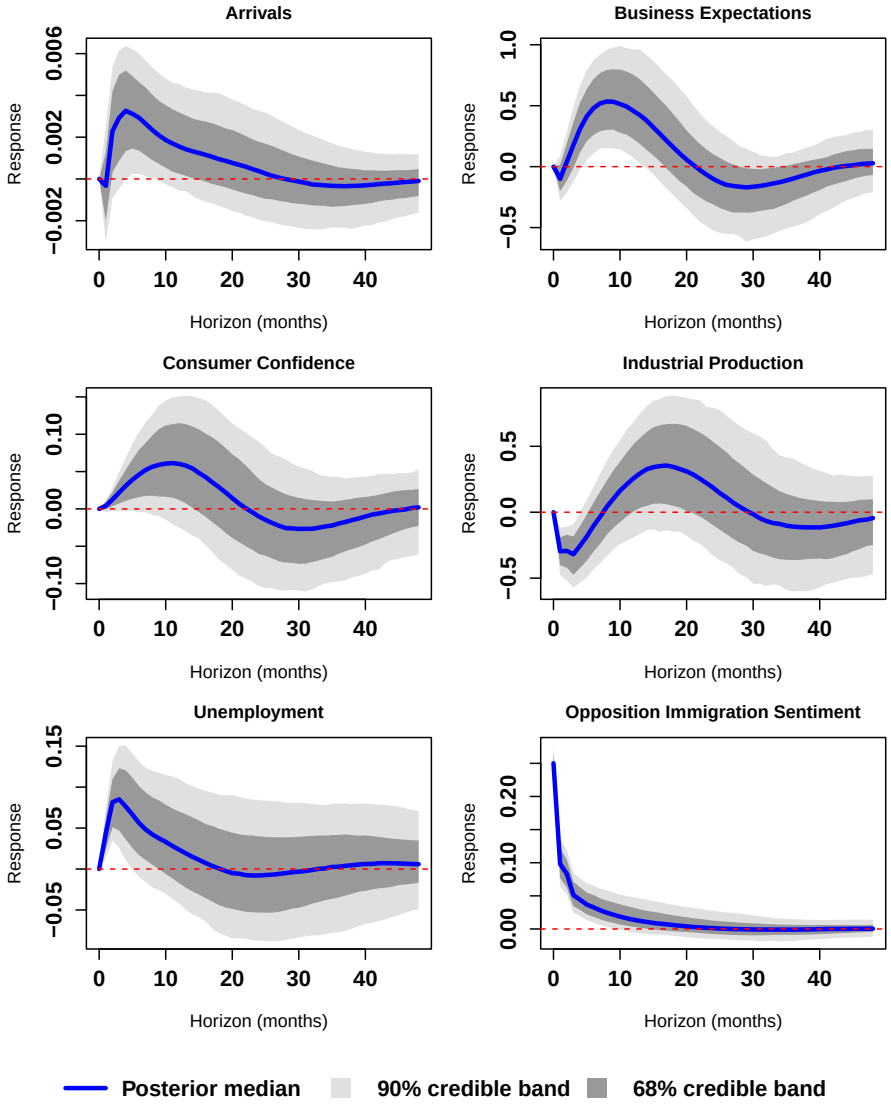


Figure 27: Robustness to the sentiment measure: impulse responses to a one-standard-deviation opposition immigration sentiment shock, with sentiment scored by GPT-4.1-nano instead of the doc2vec-based measure, new data panel, 2007M5–2019M10 ($n = 150$, $p = 2$ lags). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

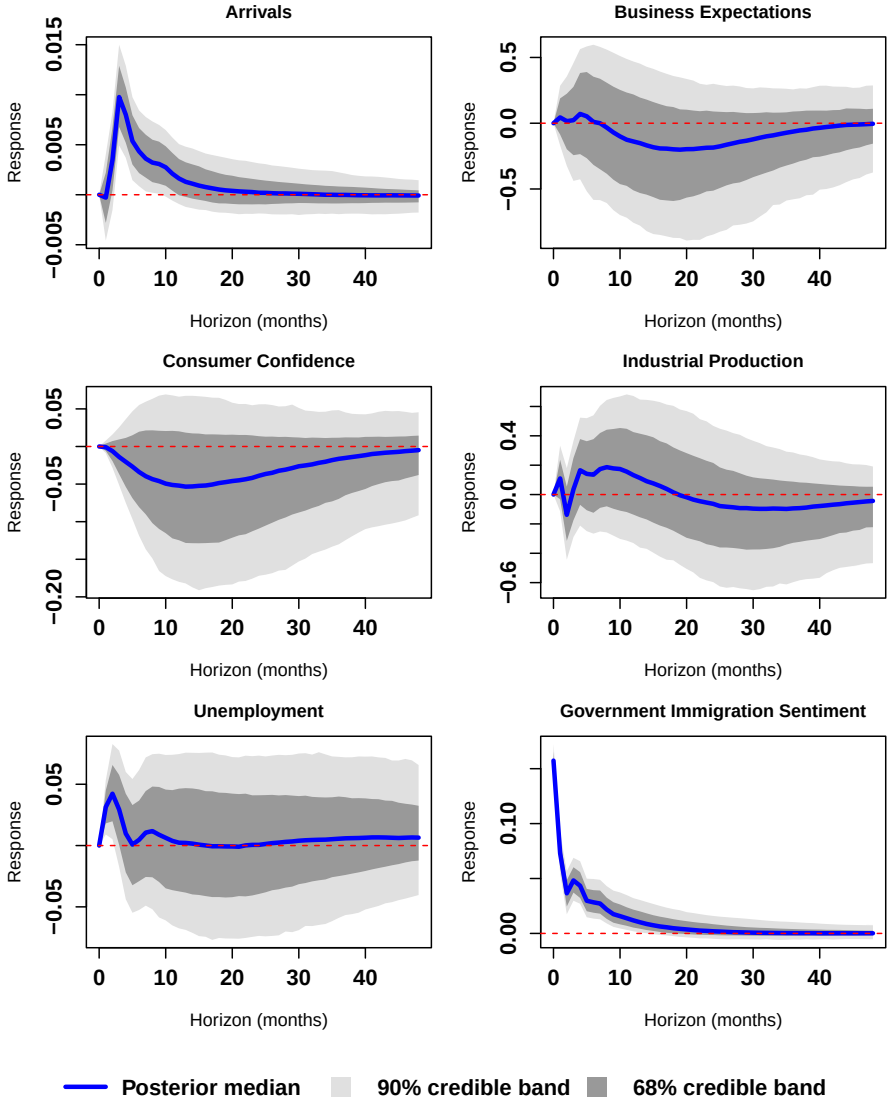


Figure 28: Robustness to the sentiment measure: impulse responses to a one-standard-deviation government immigration sentiment shock, with sentiment scored by GPT-4.1-nano instead of the doc2vec-based measure, new data panel, 2007M5–2025M6 ($n = 218$, $p = 3$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

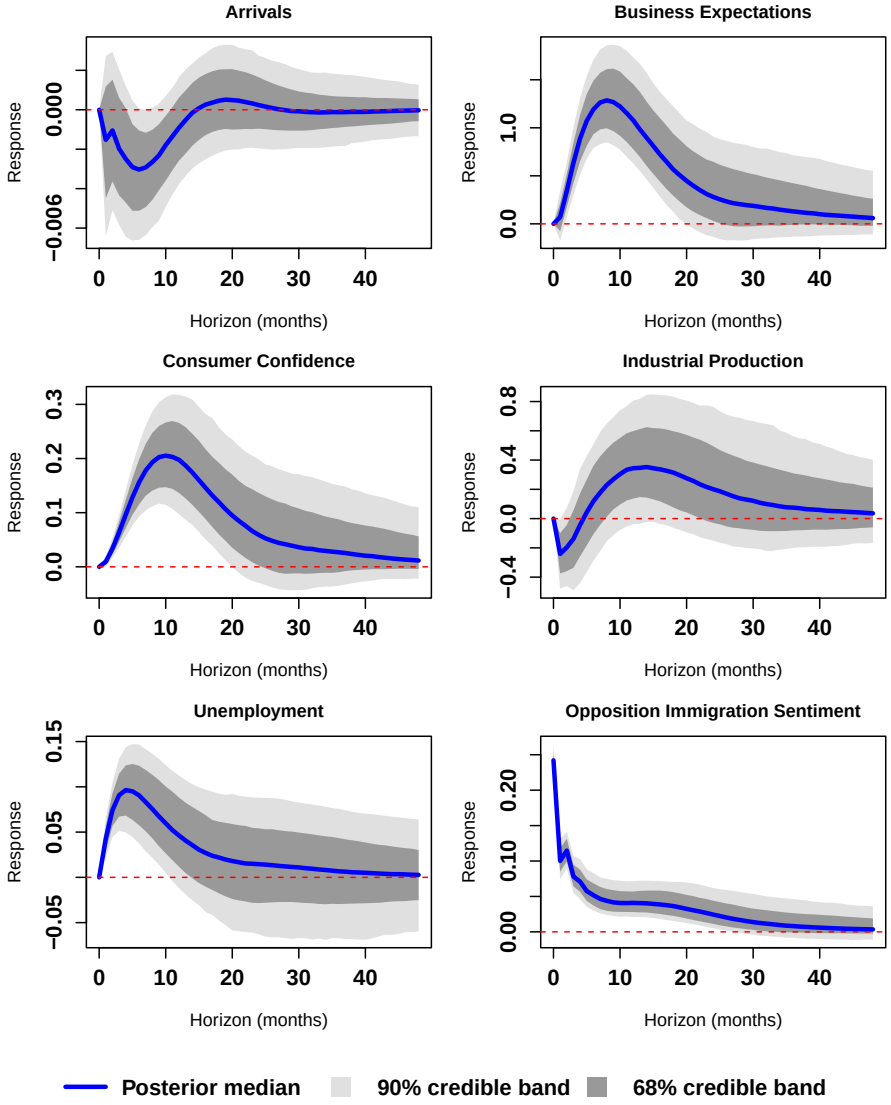


Figure 29: Robustness to the sentiment measure: impulse responses to a one-standard-deviation opposition immigration sentiment shock, with sentiment scored by GPT-4.1-nano instead of the doc2vec-based measure, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

5 Conclusion

This thesis asked whether immigration-related political sentiment in the German Bundestag carries information about later macroeconomic and migration dynamics. To answer this, the analysis combined natural language processing with a Bayesian vector autoregressive (BVAR) framework. The empirical strategy first reproduced the macroeconomic setting of Maffei-Faccioli and Vella (2021) and then extended it by adding a time-varying measure of parliamentary immigration sentiment, separately for government and opposition speakers. The results were tested across sample periods, recursive orderings, and an independent sentiment measure based on a large language model (LLM).

A central methodological contribution is the construction of a dynamic immigration sentiment measure from Bundestag speeches. The approach combines manually selected seed terms with Word2Vec-based expansion, PCA-guided refinement, and rolling-window Doc2Vec representations, which let the semantic representation of immigration-related language evolve over time without leaking future information into the measure. The comparison with the LLM-based measure also shows the limits of this approach. The Doc2Vec index captures whether speakers use welcoming or restrictive vocabulary, but not whether they endorse or criticise it, whereas the LLM classifies stance in context and reproduces the party patterns documented by Kostikova et al. (2024).

The sentiment series is associated with macroeconomic variables, and the relationship runs in both directions. Under both measures, a shift toward a more welcoming opposition stance in 2007–2025 is followed by a rise in business expectations and consumer confidence and by a rise in the unemployment rate of about 0.08 percentage points, the most stable result across all specifications. Conversely, unemployment shocks account for about 20% of the forecast error variance of opposition sentiment. Realised migration behaves differently: an arrivals shock lowers unemployment, as in Maffei-Faccioli and Vella (2021), so the sentiment shock captures information that differs from the migration flow itself. Government sentiment carries less information, and the responses are weaker for 2007–2019, which may reflect the smaller sample.

These results are dynamic associations, not causal effects. The rise in unemployment may partly reflect the delayed registration of refugees in the unemployment statistics

rather than a direct effect of rhetoric, and the opposition results depend on which parties form the opposition. Future studies could therefore separate the unemployment rates of natives and foreigners, validate the LLM classification against human annotations, and extend the analysis to other European parliaments to see whether the link between political immigration discourse and macroeconomic dynamics is specific to Germany.

Overall, the findings show that political discourse can be integrated into quantitative macroeconomic analysis in a systematic way, while also highlighting the methodological challenges of measuring political sentiment from text. The main contribution of this thesis is therefore not to establish a simple link between immigration rhetoric and economic outcomes, but to build and test an empirical framework through which such relationships can be measured, compared, and studied more rigorously.

Appendix

6 Unit Root and Cointegration Tables

Table 6: KPSS tests for level stationarity

Variable	KPSS statistic	Truncation lag	p-value	Decision
Sentiment government	3.077	5	≤ 0.01	Non-stationary
Sentiment opposition	5.187	5	≤ 0.01	Non-stationary
log(Consumer Confidence) $\times$ 100	0.770	4	≤ 0.01	Non-stationary
log(Business Expectation) $\times$ 100	1.330	4	≤ 0.01	Non-stationary
log(Industrial Production) $\times$ 100	0.689	4	0.0145	Stationary at 1%
Unemployment rate	3.423	4	≤ 0.01	Non-stationary
Arrivals	0.984	4	≤ 0.01	Non-stationary

Notes: The null hypothesis is level stationarity. Critical values: 0.463 (5%) and 0.739 (1%). p-values of 0.01 are truncated by the `tseries` implementation, so the true p-value is smaller. Industrial production is rejected at the 5% level but not at the 1% level.

Table 7: Johansen trace tests: selected cointegration rank r

Sample	Sentiment	n	K	r (1%)	r (5%)	r (10%)
2007–2025	Overall	218	3	3	4	5
2007–2019	Government	150	2	3	4	5
2007–2019	Opposition	150	2	2	3	4[a]
2007–2019	Overall	150	2	2	4	5
2007–2014	Government	92	2	3	3	4
2007–2014	Opposition	92	3	2	3[b]	4
2007–2014	Overall	92	2	3	3	3

Notes: Trace test with an unrestricted constant in the cointegrating relation (`ecdet = "const"`); r is determined sequentially from $r = 0$. K is the lag order of the VAR in levels; the six variables are Arrivals, Business Expectation, Consumer Confidence, Industrial Production, Unemployment and one sentiment series.

[a] Borderline: the test statistic (17.85) equals the 10% critical value.

[b] Borderline: the test statistic (34.84) is marginally below the 5% critical value (34.91).

7 Additional Plots

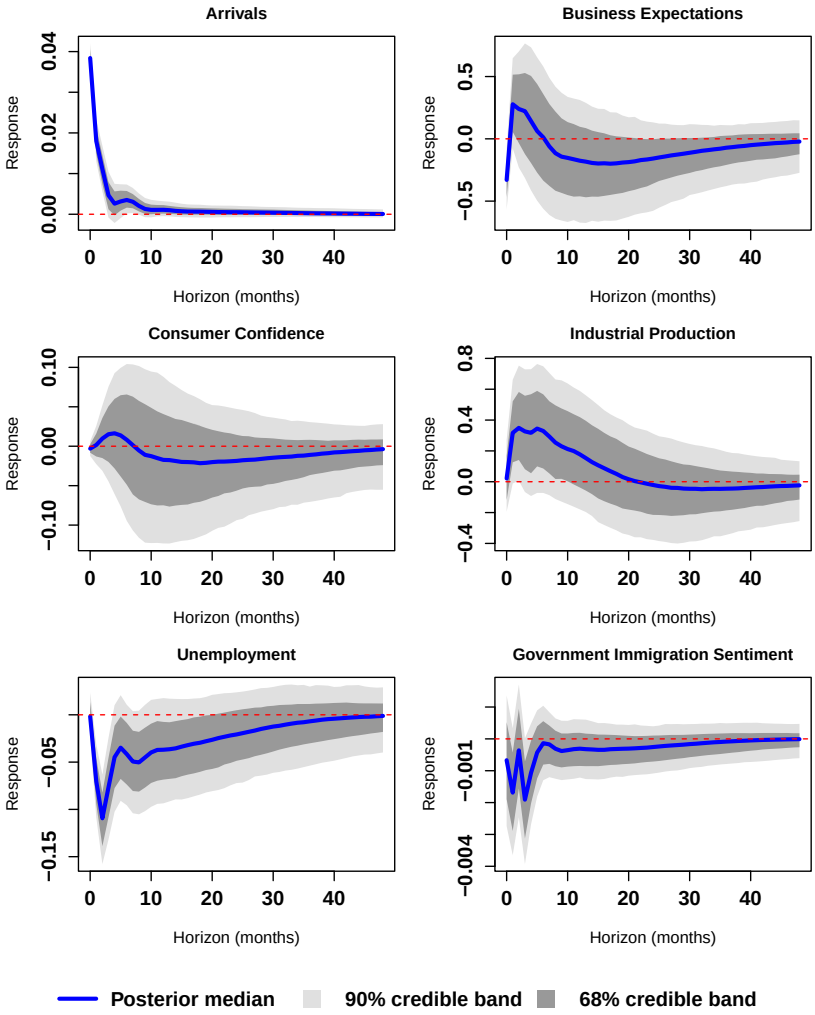


Figure 30: Impulse responses to a one-standard-deviation arrivals shock, new data panel, 2007M5–2025M6 ($n = 218$, $p = 3$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (variable ordering: arrivals, business expectations, consumer confidence, industrial production, unemployment, government immigration sentiment). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

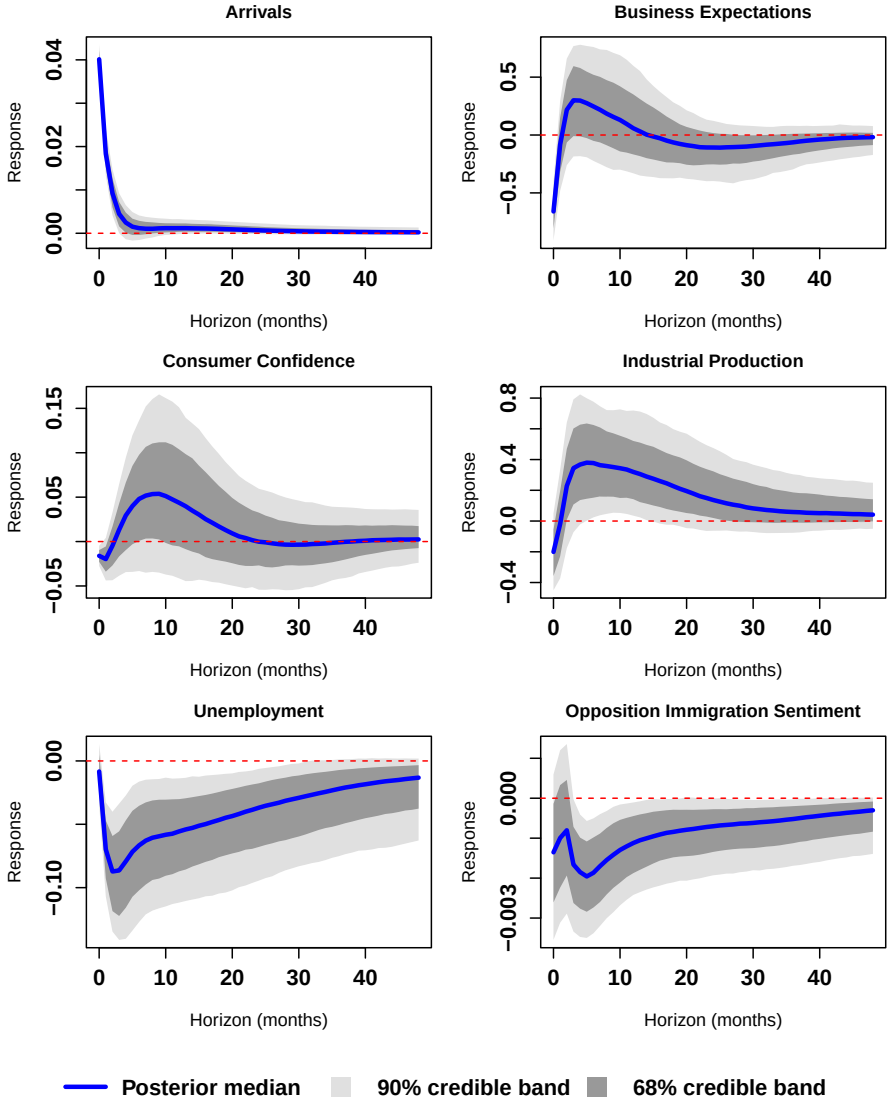


Figure 31: Impulse responses to a one-standard-deviation arrivals shock, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bayesian VAR with a diffuse prior and Cholesky identification (variable ordering: arrivals, business expectations, consumer confidence, industrial production, unemployment, opposition immigration sentiment). Lines show the posterior median; shaded areas show 68% and 90% credible bands.

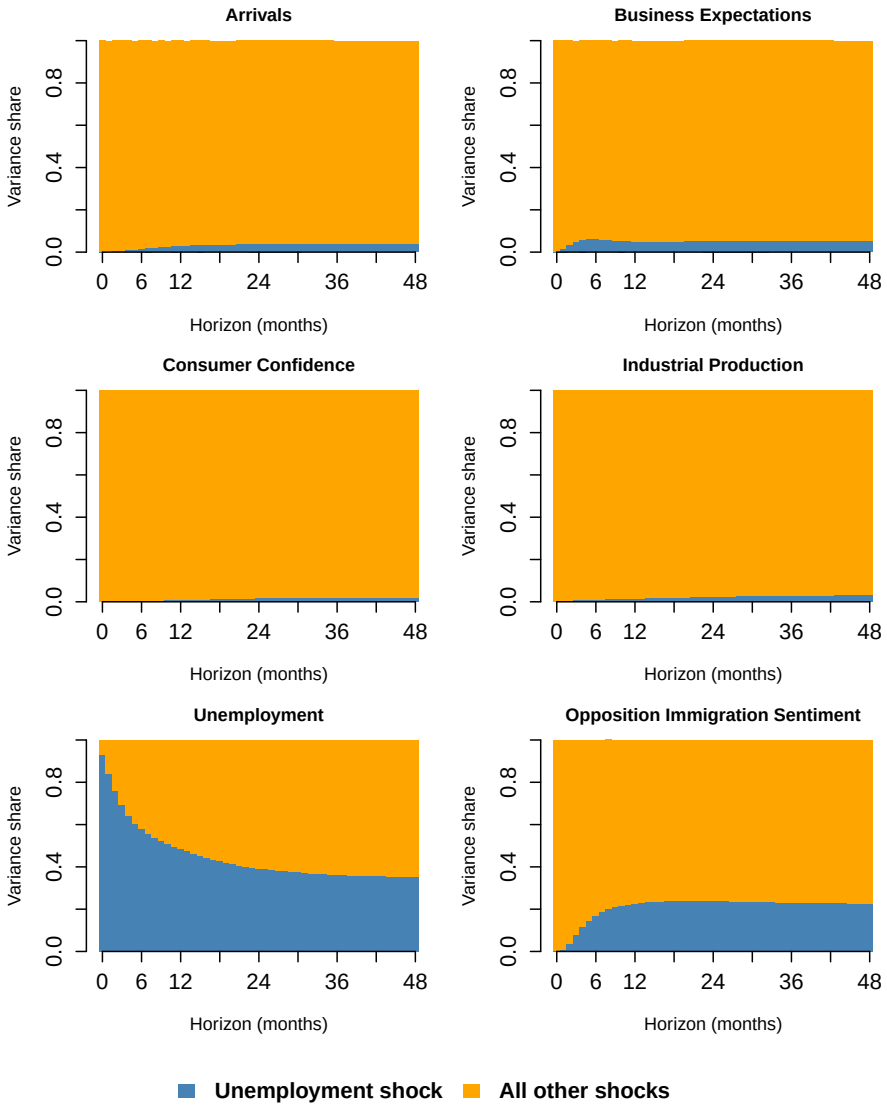


Figure 32: Forecast error variance decomposition for the model with opposition immigration sentiment, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bars show the posterior median share of each variable's forecast error variance attributable to the unemployment shock (blue) and to all other shocks (orange). Bayesian VAR with a diffuse prior and Cholesky identification (variable ordering: arrivals, business expectations, consumer confidence, industrial production, unemployment, opposition immigration sentiment); horizon in months.

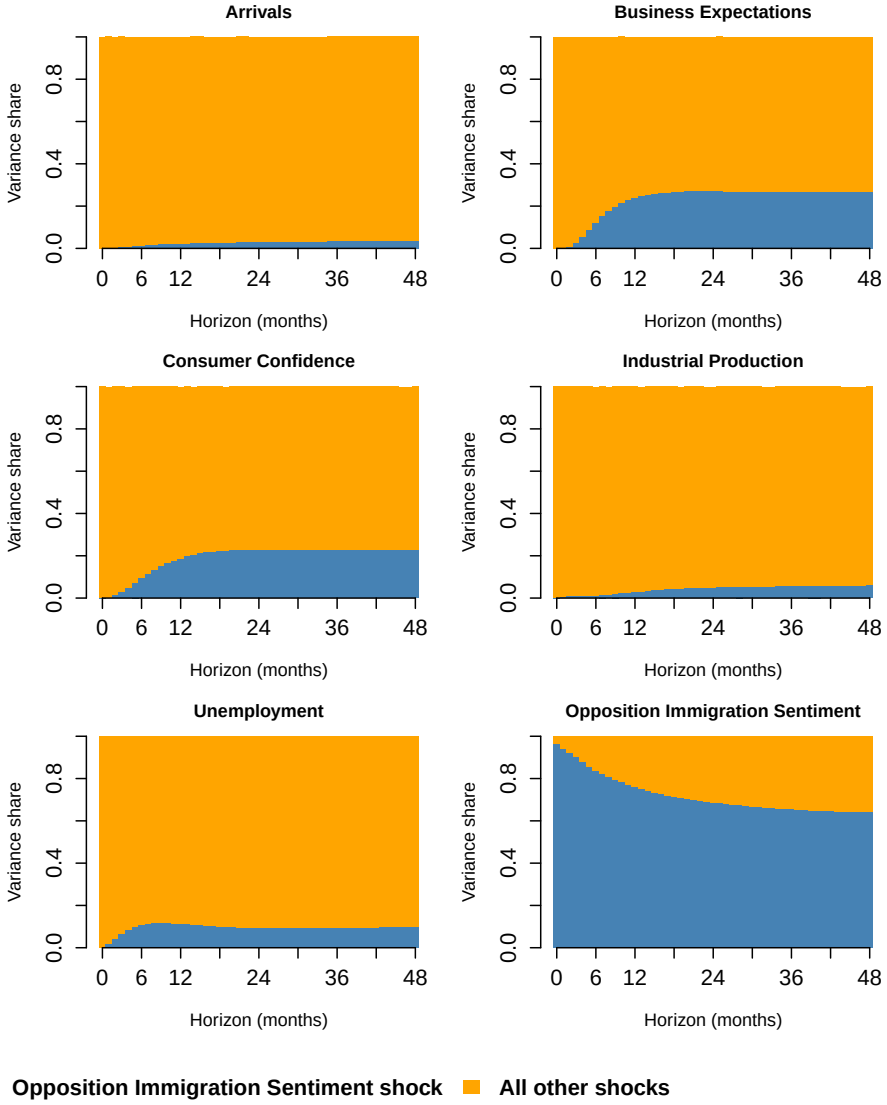


Figure 33: Forecast error variance decomposition for the model with sentiment scored by GPT-4.1-nano, new data panel, 2007M5–2025M6 ($n = 218$, $p = 2$ lags), including a COVID dummy (2020M3–2021M10). Bars show the posterior median share of each variable's forecast error variance attributable to the opposition immigration sentiment shock (blue) and to all other shocks (orange). Bayesian VAR with a diffuse prior and Cholesky identification (sentiment ordered last); horizon in months.

8 Party-Level Sentiment Statistics

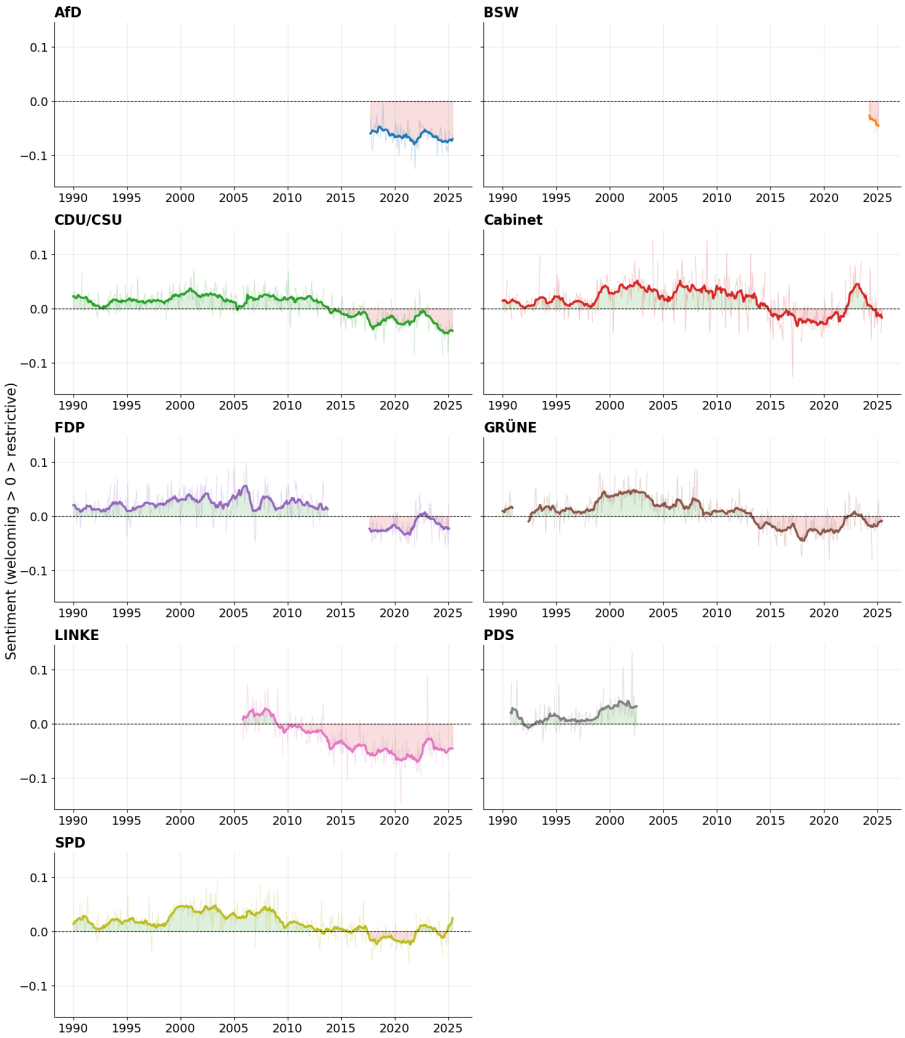


Figure 34: Evolution of Immigration Sentiment by Political Party

Notes: The figure presents the evolution of the estimated immigration sentiment scores for speeches by political party over the sample period. The horizontal red dashed line represents zero sentiment. Positive values indicate relatively pro-immigration sentiment, whereas negative values indicate relatively restrictive sentiment. The figure illustrates differences in the level and variability of immigration-related sentiment across parties and over time.

9 LLM Sentiment Classification Prompt

Sentiment classification via GPT-4.1-nano was performed using the prompt template below, generated programmatically for each speech excerpt. The five solidarity and five anti-solidarity subtypes, defined in Table 8, are inserted into the template at run-time.

Table 8: Solidarity and anti-solidarity subtypes used in prompt construction.

Subtype	**Definition**
Solidarity	
Group-based	Migrants are recognized as a legitimate group with equal rights, interests, or claims.
Exchange-based	Support is based on reciprocity: an individual migrant contributes or fulfills obligations and therefore deserves support or rights.
Economic-utility	Migration is presented as beneficial or necessary for the German economy or society (e.g., labor shortages, demographics, growth, taxes, skilled labor).
Compassionate	Migrants/refugees are portrayed as vulnerable or suffering and therefore deserving protection, sympathy, or support.
Empathic	Migrants' identity, dignity, culture, autonomy, or self-determination is explicitly recognized or respected.
Anti-solidarity	
Group-based	Migrants are portrayed as an out-group or as having fewer rights than natives; native interests are prioritized.
Exchange-based	An individual migrant is portrayed as taking more than they contribute or failing to fulfill obligations.
Economic-burden	Migration is presented as economically harmful or costly (e.g., welfare costs, fiscal burden, wage competition, pressure on resources).
Compassionate	Migrants' need for protection or humanitarian support is dismissed or portrayed as undeserving.
Empathic	Migrants' identity, dignity, culture, autonomy, or self-determination is denied or rejected.

Listing 1: Prompt template passed to the LLM for each speech excerpt.

```
Classify the following German parliamentary speech excerpt
    according to its
stance toward migrants, refugees, asylum seekers, or immigrants.

LABELS
------
SOLIDARITY = supportive, accepting, protective, inclusive, or
    sympathetic.
ANTI-SOLIDARITY = opposing, excluding, rejecting, hostile, or
    unwilling to support.
MIXED = both meaningful solidarity and anti-solidarity.
NONE = no clear stance toward migrants.

RULES
-----
- Migration must be the target. Mentioning migration alone is
    not enough.
- Neutral facts, statistics, questions, or policy descriptions =
     NONE.
- Criticism/support of government or migration policy is not
    automatically
anti/pro-migration; classify only the stance toward migrants.
- Do not classify unrelated groups or historical German
    expellees as migrants.
- If the evidence is unclear, choose NONE.
- If SOLIDARITY or ANTI-SOLIDARITY, choose exactly ONE subtype.
- If MIXED or NONE, subtype = NONE.

SOLIDARITY SUBTYPES
-------------------
[inserted from Table~\ref{tab:solidarity_subtypes}]

ANTI-SOLIDARITY SUBTYPES
------------------------
[inserted from Table~\ref{tab:solidarity_subtypes}]

TEXT
----
```

```
**<speech excerpt>**

OUTPUT FORMAT
-------------

Return EXACTLY two lines and nothing else:

Justification: <one short sentence explaining the decision>
Label: <HIGH-LEVEL>[-<SUBTYPE>]

VALID LABELS
------------

SOLIDARITY-GROUP-BASED
SOLIDARITY-EXCHANGE-BASED
SOLIDARITY-ECONOMIC-UTILITY
SOLIDARITY-COMPASSIONATE
SOLIDARITY-EMPATHIC

ANTI-SOLIDARITY-GROUP-BASED
ANTI-SOLIDARITY-EXCHANGE-BASED
ANTI-SOLIDARITY-ECONOMIC-BURDEN
ANTI-SOLIDARITY-COMPASSIONATE
ANTI-SOLIDARITY-EMPATHIC

MIXED
NONE
```

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Declaration of originality

I hereby declare that I have written this Masterthesis independently and without using other than the indicated sources and aids and that I have identified the passages taken verbatim or in content from the used sources as such. The work in the same or similar form has not been submitted to any other examination authority.

Linden, 02.11.2026

Esdras Shaa Prombo